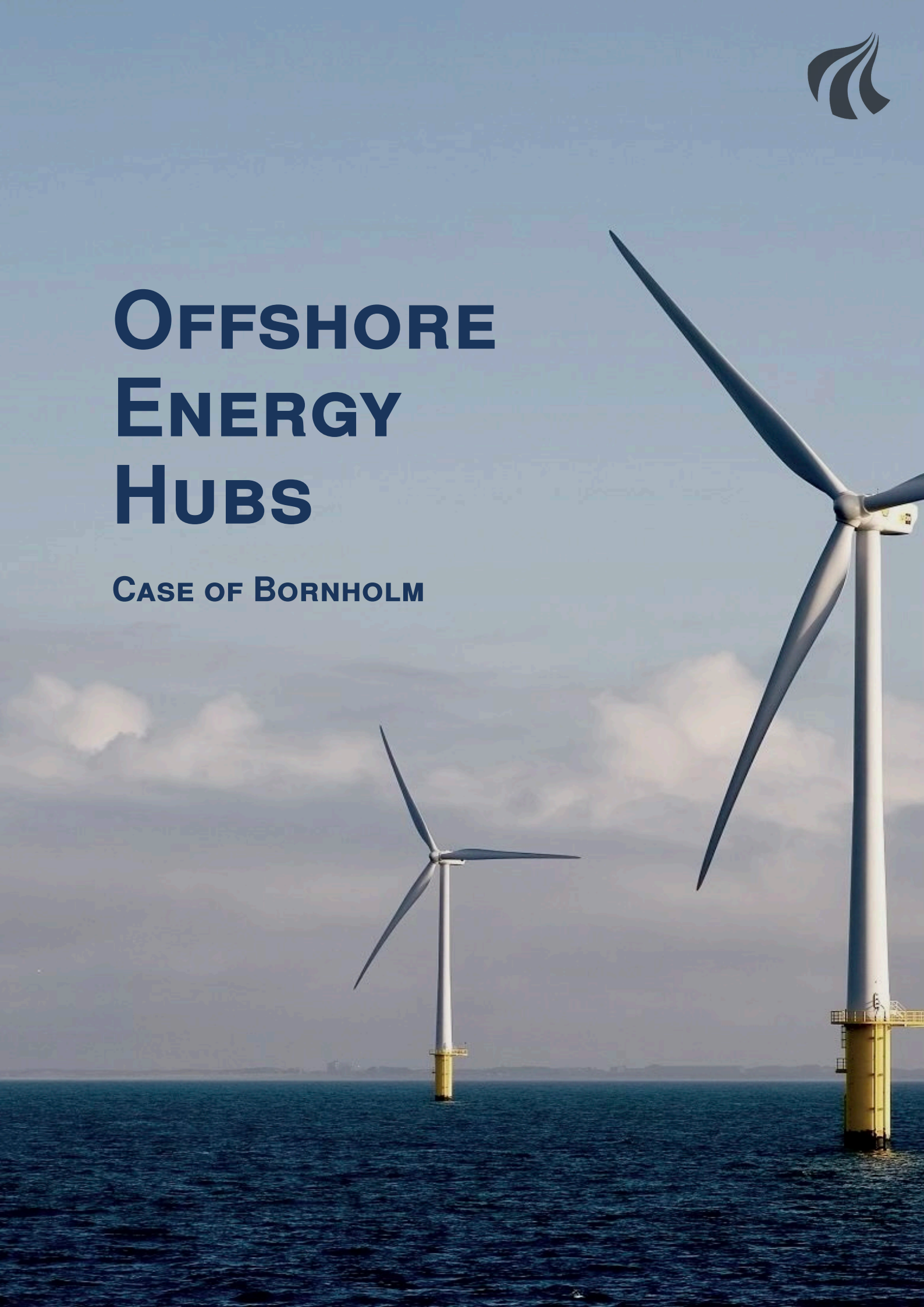
Offshore Energy Hubs

Case of Bornholm



First Semester Project - Title Page

Project title	Offshore Energy Hubs – Case of Bornholm
Education	MSc in Urban, Energy and Environmental Planning
Specialisation	Sustainable Energy Planning and Management
Hand-in date	8th January 2026
Supervisor	Alisson Aparecido Vitoriano Julio
Group members	Julia Wiktoria Ryfa 20251549 Raphaël Dimitri Cabay 20252384 Marco Schumacher 20251941 Marcelina Majchrzak 20252412

Abstract

This project assesses Bornholm's potential role as an offshore energy hub by examining four development scenarios, ranging from the approved baseline of 3,2 GW offshore wind and grid connections to Denmark and Germany to expanded wind power generation combined with ammonia-based Power-to-X production. Using WindPro and EnergyPro, the scenarios are modelled to evaluate technical performance, energy flows and the economic feasibility of different technological configurations, including electrolyser and ammonia synthesis capacities, under 2024 electricity price conditions. The analysis shows that all scenarios are technically feasible and that wind-only configurations exhibit lower levelised costs of electricity, while Power-to-X integration mainly acts as a flexibility option that can enhance system value under favourable ammonia and electricity price developments. Overall, the project concludes that the Bornholm Energy Island is a viable and strategically important investment, where immediate economic benefits are driven primarily by offshore wind, while Power-to-X infrastructure provides long-term optionality in line with European Union climate and energy security objectives.

Use of AI Tools

For this report, Artificial Intelligence (AI) tools were used only as support for the research and writing process, and not to generate the actual text included in the document. The authors are solely responsible for all content, argumentation, and wording.

AI was used to:

- Help find relevant literature and refine search terms, with all sources independently checked by the authors.
- Summarise and clarify academic texts, with information verified against the original sources.
- Suggest language and grammar improvements to drafts, which were manually reviewed and edited before inclusion.

AI was not used to draft paragraphs or sections of this report, and no AI system is considered an author or a citable scholarly source. Any suggestions from AI tools were critically evaluated and only incorporated after verification and deliberate author judgment.

List of Abbreviations

AAU – Aalborg University

AI – Artificial Intelligence

BEI – Bornholm Energy Island

DK2 – East Denmark (Zealand)

DE – Germany

DTU – Technical University of Denmark

EU – European Union

GDP – Gross Domestic Product

GHG – Greenhouse gases

GIS – Geographic Information System

HVDC – High-Voltage Direct Current

LCOE – Levelised Cost of Energy

NGOs - Non-Governmental Organisations

NPV – Net Present Value

OEH – Offshore Energy Hub

O&M – Operation and maintenance

PEMEC – Proton Exchange Membrane Electrolytic Cells

PL – Poland

PV – Present Value

PtX – Power-to-X

RES – Renewable Energy Sources

SDU – University of Southern Denmark

SE – Sweden

TSO – Transmission System Operator

Contents

1 Introduction - problem analysis **1**

1.1 The status quo of global energy 1

1.2 EU climate policies 2

1.3 Smart energy systems 4

1.4 Offshore energy hubs 5

1.5 Danish climate agreement 6

2 Problem statement **8**

3 Research design **10**

3.1 Structure and content 11

3.2 Delimitations 11

4 Literature review **13**

5 Methodology **17**

5.1 Case study - Bornholm Energy Island 18

5.2 Interview and conference 21

5.3 EnergyPro 23

5.4 WindPro 24

5.5 Scenarios development 24

6 Technical and economic assumptions 27

6.1 Technical analysis 27

6.1.1 Offshore wind farm 27

6.1.2 Cable connections 27

6.1.3 Hydrogen 28

6.1.4 PtX 29

6.2 EnergyPro models 31

6.2.1 Scenario 0 - Baseline 31

6.2.2 Scenario 1 33

6.2.3 Scenario 2 34

6.2.4 Scenario 3 35

7 Analysis of the results 36

7.1 Comparison 36

7.1.1 Annual energy conversion summary 36

7.1.2 Annual economic summary 39

7.1.3 Time perspective 42

7.2 Sensitivity analysis 44

7.2.1 Electricity prices - disparity 44

7.2.2 Electricity prices - development 47

7.2.3 Cost of CO_2 emissions 48

7.2.4 PtX 49

7.2.5 Losses in the HVDC cables 51

8 Discussion **52**

8.1 Technical challenges 52

8.2 Cost-effectiveness 53

8.3 Risks and uncertainties 54

8.3.1 Sensitivity analysis 54

8.3.2 Stakeholders and community acceptance 56

9 Conclusion **62**

10 Appendix **70**

10.1 Interview IVA 70

10.2 Power to X 75

10.2.1 PEMEC electrolysis 75

List of Figures

1 World's total energy supply (source: [5]) 2

2 The concept of a smart energy system involving different inputs and outputs (source: own work) 5

3 Simplified concept of an energy hub (source: own work) 6

4 Logical path (source: own work) 10

5 Research question (source: own work) 17

6 Bornholm Island, situated off the coast of Denmark and other European countries (source: own work) 18

7 Location of the planned wind farms near Bornholm (source: own work) 20

8 The concept of Bornholm Energy Island (source: own work inspired by [31]) 21

9 Scenario space (source: own work inspired by [33]) 26

10 Power Curve of the wind turbine IEA 15 MW 240 RWT (source: [35]) 27

11 Power to Ammonia - simplified schematic (source: own work) . . . 30

12 EnergyPro - site overview (source: screenshot from EnergyPro) . . . 31

13 EnergyPro - site overview, scenarios 0 and 1 (source: screenshot from EnergyPro) 33

14 EnergyPro - site overview, scenarios 2 and 3 (source: screenshot from EnergyPro) 35

15 Consumption and transmission of energy - 2024 perspective (source: own work) 36

16 Consumption and transmission of energy - 2024 perspective (source: own work) 37

17 Consumption of nitrogen and production of ammonia - 2024 perspective (source: own work) 38

18 Results of simulation - 2024 perspective (source: own work) 38

19 Yearly outflow and revenue - 2024 perspective (source: own work) 41

20 Capital, variable and external costs - 2024-2054 perspective (source: own work) 42

21 Evaluation of Offshore Energy Hub power exchanges and price signals between DK2 and Germany for low and high wind production conditions - time period example. (source: own work) . . . 45

22 Evaluation of Offshore Energy Hub power exchanges and price signals between DK2 and Germany for low and high wind production conditions and ammonia production - time period example. (source: own work) 46

23 Change of electricity prices throughout the years for Eastern Denmark and Germany (source: own work) 47

24 Consumption and transmission of produced energy - scenario 3 in different years (source: own work) 47

25 Stakeholders matrix (source: own work) 57

26 Schematic of a PEMEC cell (source: [38]) 75

List of Tables

1 Variable costs (source: own work) 40

2 External costs (source: own work) 40

3 Capital costs (source: own work) 40

4 Annual revenue (source: own work) 41

5 Yearly balance - 2024 perspective (source: own work) 41

6 LCOE - 2024-2054 time horizon with 5 % discount rate (source: own work) 43

7 NPV - 2024-2054 time horizon with 5 % discount rate (source: own work) 44

8 Change of NPV - 2024-2054 time horizon with 5 % discount rate - based on CO_2 prices (source: own work) 48

9 Net Present Value for different PtX capacities (2024–2054, 5% discount rate) (source: own work) 50

10 Net Present Value for different ammonia prices (2024–2054, 5 % discount rate) (source: own work) 50

1 Introduction - problem analysis

1.1 The status quo of global energy

Given today's uncertainty and unstable geopolitical situation, it is imperative to recognise the consequences of inaction on climate change and set explicit goals within defined timeframes to ensure energy security. Sustainable development should be seen as an ongoing process, extending beyond reducing energy consumption and losses to also consider the future of the economy and energy sector.

Global energy demand has grown by an average of 1,4 % per year over the last decade [1], while energy efficiency improvement - measured as total final energy consumption per unit of the Gross Domestic Product (GDP) - remains below par, averaging just 1,5 % per year between 2019 and 2024, down from nearly 2 % per year over the previous 10 years. This weakness underpins the continued steady growth in fossil fuel consumption despite the rapid increase in the share of low-carbon energy [2]. When discussing the increasing demand for energy, it is crucial to consider the construction of data centres, which have become a necessity due to growing interest in AI [3, 4].

Fossil fuels, as the primary source of energy (see Figure 1), play a dominant role in global energy systems. However, as we are talking about a non-renewable resource, we must pay attention to the depletion of them. The data is typically presented as a reserve-to-production ratio, which measures the number of years remaining until 'proven reserves' (reserves that could be extracted in the future under current economic and technological conditions) are depleted. According to the Statistical Review of World Energy from 2025, created by the Energy Institute, coal reserves will last for 139 more years, oil reserves for 56 more years, and gas reserves for just 49 more years.

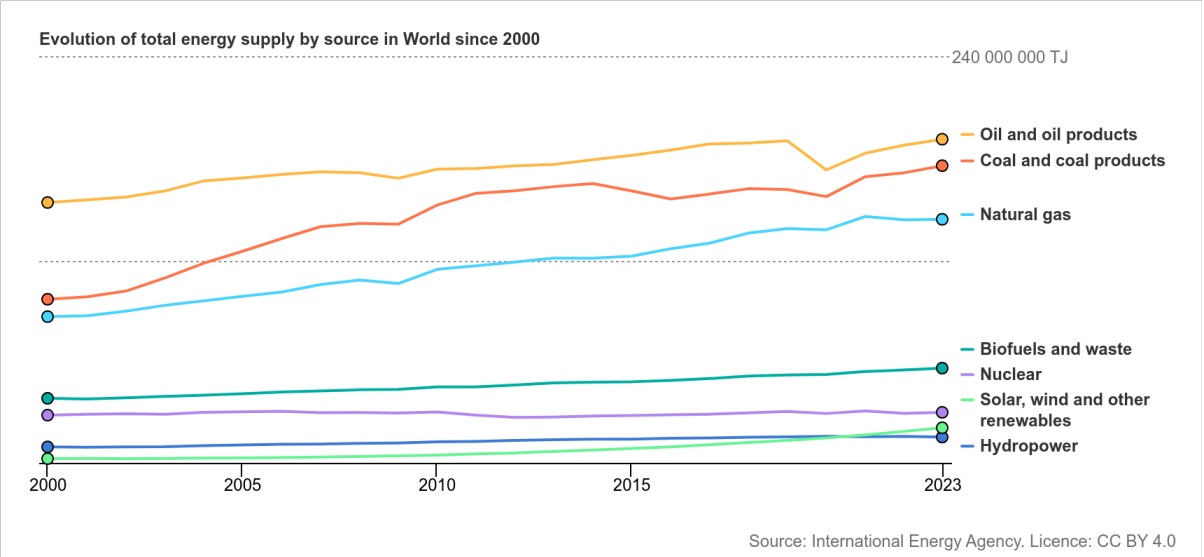


Figure 1: World's total energy supply (source: [5])

As a more electrified and renewable-rich energy system will play a key role in decarbonising our energy systems in the coming decades, the world must rapidly transition to the technologies that appear most promising in transforming our energy mix. Although modern technologies are expected to enable a significant decline in energy demand in a sustainable manner, it is necessary to develop energy efficiency policies that ensure affordability and energy security. This requires innovative approaches at both the European and national levels [6].

1.2 EU CLIMATE POLICIES

The European Union (EU), widely regarded as a leader in socio-economic development, often sets standards for the world. As one of the world's largest economies and a major emitter, the EU has taken a strong responsibility for addressing climate change. It has taken the lead by implementing comprehensive policy frameworks to mitigate and adapt to climate change while promoting sustainable development. These aim to transform its economy into one that is low-carbon, resource-efficient, and socially inclusive.

The need for climate action was first officially recognised in the 2016 Paris Agreement, signed by 195 countries. This document aims to limit the global rise in temperature to "well below 2 °C above pre-industrial levels", also pursuing

efforts to limit the rise to 1,5 °C. In response, the EU launched the European Green Deal in 2019 to achieve climate neutrality by 2050. It focuses on decoupling growth from emissions, protecting biodiversity, supporting a circular economy, and ensuring a fair transition. The European Climate Law, adopted in 2021, made these goals legally binding and set two main milestones: a 55 % reduction in emissions by 2030 (compared to 1990 levels) and climate neutrality by 2050 [7].

The European Climate Law is the foundation for all subsequent EU climate policies and legislative changes. According to Article 2, EU-wide emissions and removals must be balanced by 2050, with at least a 55 % reduction by 2030. The Renewable Energy Directive requires Member States to reach a 42,5 % share of renewables by 2030, with an ambition of 45 % [8]. The Energy Efficiency Directive calls for an 11,7 % cut in final energy use compared to 2030 forecasts, while the REPowerEU Plan (2022) supports all this by speeding up the use of renewables and aiming to reduce reliance on Russian gas [9].

Although EU frameworks establish common objectives, Member States adapt them to their own specific national conditions. For example, Germany aims for climate neutrality by 2045 under its Federal Climate Change Act (2021), planning to phase out coal by 2038 and expand renewables and hydrogen use [10]. Denmark, a Northern European frontrunner in energy transition, has a goal to remove fossil fuels from electricity and heating by 2030. It is also a global leader in offshore wind, developing "energy islands" in the North and Baltic Seas that promote regional cooperation [11].

In summary, the development of EU climate policy, from the Paris Agreement to the Green Deal and Climate Law, shows a firm dedication to achieving climate neutrality. Through binding targets, financial tools, and national initiatives, the EU has become a global climate policy leader, proving that sustainability and prosperity can go hand in hand. As the European Commission stated in 2019, the European Green Deal is "a new growth strategy that gives more back to the planet than it takes away" [7].

1.3 Smart energy systems

Based on the goals established in official documents [12], it can be said that Europe's energy will largely depend on renewable sources. Unlike conventional sources, which are flexible in their use and whose power depends on consumption, renewable energy sources (RES) are subject to natural conditions. This means the entire energy system needs to be adapted to manage energy as efficiently as possible.

Variable input is just one of the challenges facing current and future energy systems. Currently, most energy sources are non-renewable and must be able to work alongside an increasing number of renewable sources. When planning country's energy future, it is not enough to consider a single energy sector; a broader range of infrastructure must be taken into account.

Recently, the term 'smart energy systems' has become increasingly frequent in discussions about sustainable energy strategies for the future. The term 'smart energy systems' signifies a shift away from single-sector thinking towards a coherent and integrated understanding of how to design and implement the most achievable and affordable strategies for future sustainable energy systems [13]. Within a single, coherent energy system, a number of synergies between sectors can be identified, which would not be possible with an approach that only considers renewable sources or the heating sector in isolation. This radical change in the way we think about energy systems, combined with a more holistic approach, enables us to find optimal, achievable, and affordable solutions for the transition to a renewable and sustainable energy sector.

Smart energy systems focus on the proper integration of variable RES into the existing system, while considering energy-saving, -storage and -conversion technologies (see Figure 2). The main objective is to achieve optimal energy and cost efficiency. Due to the variety of energy carriers used as inputs, it is possible to provide a flexible energy supply based on demand using RES, among others.

The advantages of such advanced concepts, which enable the optimal management of various energy carriers and systems, can be utilised in the form of energy hubs. An energy hub is a unit that provides functions for the input and output, but also conversion and storage of multiple energy carriers [14].

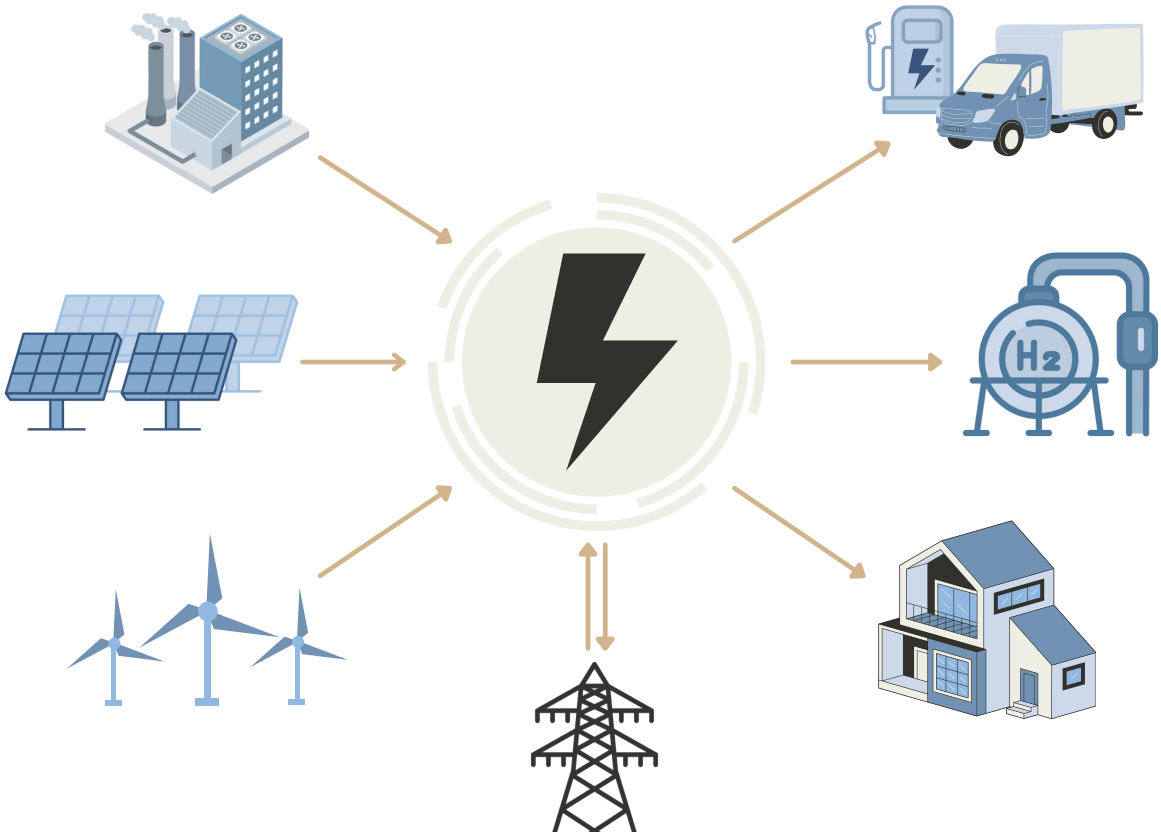


Figure 2: The concept of a smart energy system involving different inputs and outputs (source: own work)

Such a system gives us greater freedom, particularly with regard to meeting energy demand and exploiting it fully. Energy in various forms can be supplied to many consumers, optimising the utilisation of energy resources and increasing efficiency.

1.4 Offshore energy hubs

Depending on many factors, different types of technologies for energy production, conversion or storage can be used. One of the RES is wind, which can be used in wind power plants to generate electricity. Generally, the higher the wind speed, the greater the electricity production. The construction of wind turbines frequently occurs in the context of offshore wind turbine pro-

jects. The wind in this region is characterised by greater strength and regularity, which renders it a more optimal solution from an operational perspective. Consequently, the notion of an offshore energy hub (OEH) was proposed.

An OEH or energy island can be defined as a hub with surrounding offshore wind farms connected to it and transmission infrastructures between the hub and the neighbouring energy systems. Energy transferred to the land can then later be used in one of the energy systems, transformed, stored, or used for an exchange of energy between the connected energy systems/countries. Energy islands can use electricity, hydrogen, or be the first step towards creating a complete offshore energy network with multiple connected islands/hubs [15].

As OEHs could form part of a future sustainable energy system, investments in this area must be made carefully and in a cost-effective manner. Depending on the chosen location, certain constraints must be applied in order to select the most optimal solution, taking into account various aspects such as emissions, energy losses, and costs.

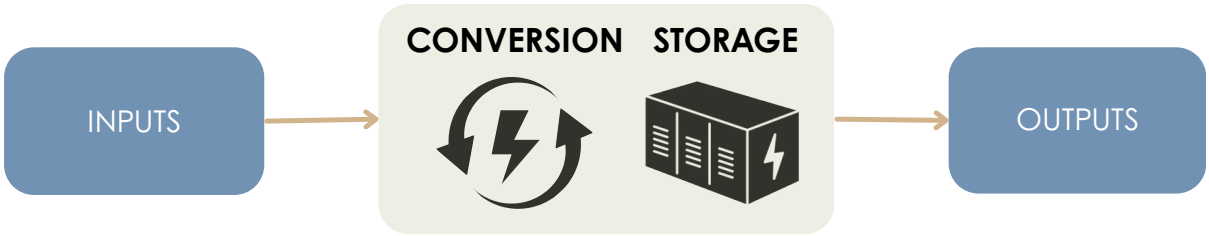


Figure 3: Simplified concept of an energy hub (source: own work)

1.5 Danish climate agreement

In 2020, the Danish Ministry of Climate, Energy and Utilities published the Danish Climate Agreement for Energy and Industry. In the response to Paris Agreement, mentioned in the subsection 1.2, Denmark aims to cut Greenhouse Gases (GHG) emissions by 70 % by 2030 compared to 1990 levels and to become climate-neutral by 2050, if not sooner [16].

A key initiative of the agreement is for Denmark to become the world's first country to enter a new phase of offshore wind expansion by shifting from in-

dividual offshore wind farms to energy islands. This transition, planned to be completed by 2030, aims to establish the energy sector of the future. This will ensure the more efficient use of non-coastal offshore wind resources and reduce the need for new land-based power infrastructure. It can support the electrification of Denmark and assist Europe's green transition through Danish exports and the integration of European electricity markets [16].

The agreement sets out plans to establish two energy islands by 2030. One would be located in the North Sea and would have an estimated capacity of 3 GW of offshore wind power, with the potential to accommodate an additional 7 GW in the long term. The other one would be located in the Baltic Sea on the Danish island of Bornholm. Up to 2 GW of offshore wind power would be connected to Denmark by this energy island [16].

As part of Denmark's ambition to lead the global green transition, Bornholm's status as an energy island is central to achieving the objectives of the Danish Climate Agreement. The project will contribute significantly to national emissions targets and serve as a test case for integrating large-scale offshore energy systems into regional and European networks. Despite the ambitious plans and political vision surrounding Bornholm Energy Island (BEI), knowledge of its potential economic and technological impacts remains limited. The following chapter therefore defines the problem statement that will guide this report, focusing on Bornholm as a case study [15].

2 Problem statement

As mentioned in section 1.4, OEHs have become central to European strategies to reduce greenhouse gas emissions and increase energy independence from fossil fuels, as well as enabling large-scale electrification. As part of its national climate plan, Denmark has introduced OEHs designed to collect and distribute electricity on a large scale. The BEI project is one of the first practical steps towards implementing this model, serving as both a transmission node and a potential foundation for future energy conversion or storage technologies.

Although the baseline project configuration has been approved, the long-term development path remains uncertain. The current configuration consists of from 3,2 GW up to 3,8 GW of offshore wind capacity and high-voltage direct current (HVDC) interconnectors to Denmark and Germany [17]. There is little analytical work on how different expansion strategies influence the technological performance of the system and the economic feasibility of investing in additional infrastructure. Although existing studies have considered Power-to-X (PtX), these have exclusively addressed the social and environmental dimensions, excluding considerations of economic viability or integration with the augmentation of wind farm capacity.

This gap highlights the need for a systematic evaluation of how BEI could develop beyond the scope of the approved project. It is essential to understand the technological configuration requirements and economic conditions under different development pathways in order to identify cost-effective and technically viable expansion strategies.

This project therefore assesses Bornholm's potential role as an OEH by examining four development scenarios, ranging from the approved baseline to expanded wind power generation linked to PtX integration. The aim is to identify the differences in system performance, infrastructure needs, and economic outcomes between these scenarios.

In light of all this, the research question is as follows:

Research Question

How do different development pathways of the BEI project affect its technological configuration and economic feasibility?

Sub-Questions

1. What are the technical challenges of establishing the BEI project examining different development scenarios?

2. What are the projected investment costs, operational expenditures, and revenue streams for each scenario, and how do they impact the overall cost-effectiveness of the BEI?

3. What are the main risks and uncertainties associated with these scenarios, and which actors are expressing the greatest concern regarding them?

3 Research design

The logical path towards finalising the research question, from broad global energy challenges to the specific Danish Bornholm context, is presented below.

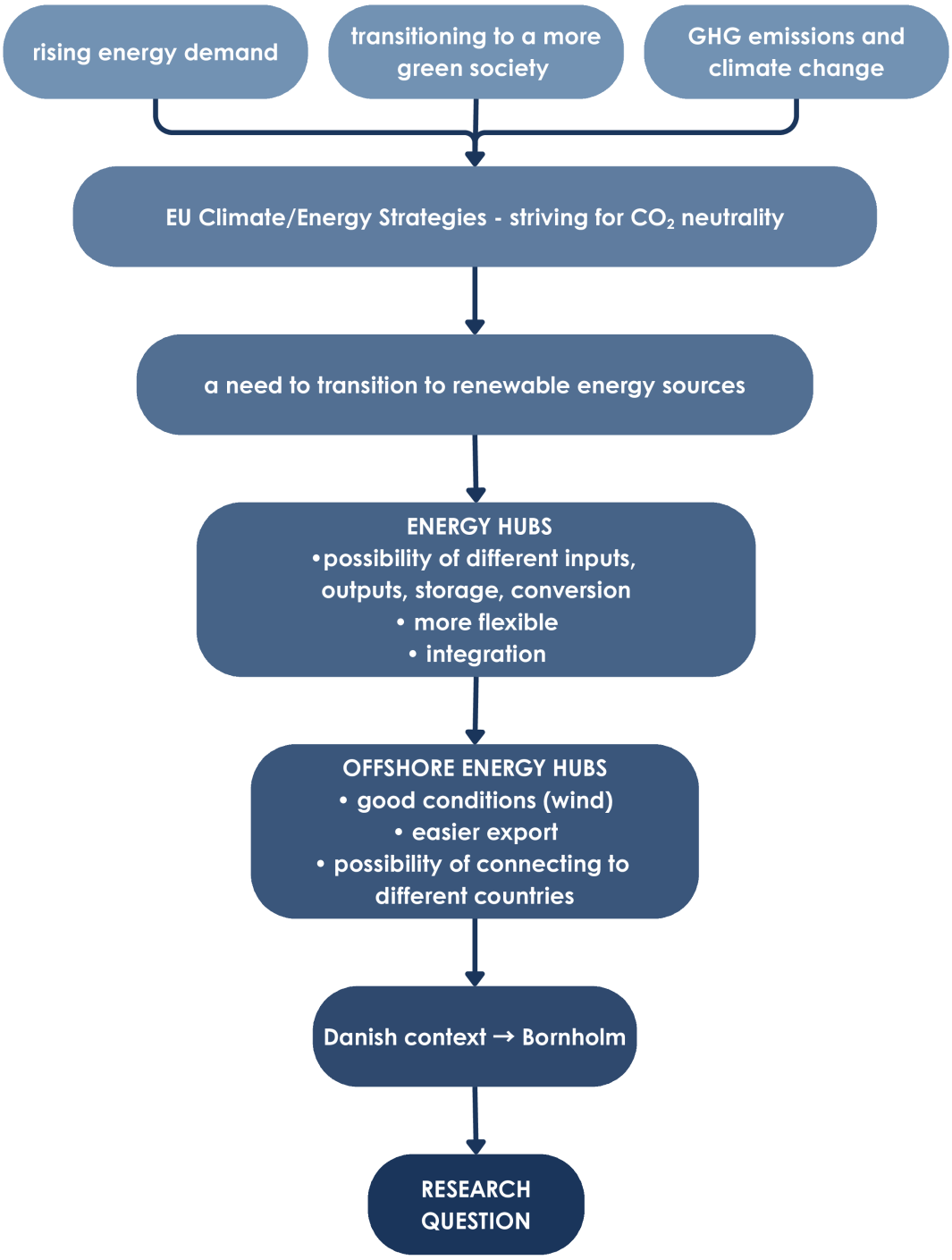


Figure 4: Logical path (source: own work)

3.1 Structure and content

In order to effectively answer the research question presented in the problem statement, a plan was established regarding the objectives, methodology, and techniques that would be used to achieve these objectives. The framework presented an overview of the primary topics that were evaluated and subsequently developed in later stages.

The initiation of this project began with a thorough analysis of the problem at hand. This analysis was then used to formulate a logical path (see Figure 4), with the initial focus being on objectives related to the effective management of growing energy demand and the depletion of traditional fuel resources. A synopsis of the EU's most significant strategies was provided, underscoring the necessity to employ an energy mix predominantly reliant on RES. The research subsequently focuses on smart energy systems and the concept of energy hubs, which, as the study will demonstrate, should serve as the fundamental framework for the future energy sector. Following the decision to concentrate on the Danish perspective, the island of Bornholm was selected as a potential location for the establishment of an OEH.

3.2 Delimitations

In order to focus the research around clear conceptual boundaries, certain limitations had to be established. The following statements were selected as guidelines for the entire report to provide direction for the delimitation process:

- The analysis of the issues highlighted the need for sustained action in many places around the world. Denmark was selected as a representative country for the report, where the possibility and will to establish an OEH was recognised.
- While the projects adopted at this stage do not result in Bornholm being recognised as an energy hub, the prospect of future projects and clear signals from key stakeholders that the OEH will be established means that

all scenarios in the report will assume that Bornholm will become an energy island.

- The primary objective of the analysis is to assess the potential for transforming Bornholm into an OEH and the consequences of achieving such a transformation. Currently approved projects are well thought out, however, it is imperative that further development of the projects is undertaken.
- The expansion under consideration is presented in the form of plausible scenarios, which are based on relevant articles concerning the future outlook of the island and an understanding of the direction of development of the energy sector in Europe.
- A comprehensive analysis was conducted, exploring various expansion prospects, including the establishment of cable connections with nearby countries, the initiation of hydrogen production, and the potential surge in demand driven by the emergence of energy-intensive industries on Bornholm. The decision was taken to restrict the scope to the expansion of offshore wind farms and the production of ammonia.
- In the calculations, all energy produced at OEH is used for transmission to Germany, Zealand or for the production of ammonia. The island's own energy demand is not addressed in the study.
- The choice to produce ammonia as opposed to other alternative fuels was motivated by the aspiration to use a zero-emission fuel, whilst acknowledging its potential as a convenient method of transporting and storing clean hydrogen.
- In accordance with the objectives of the report, it is assumed that the wind farms in question would be entirely newly constructed. This assumption provides a greater degree of flexibility in the modelling of the scenarios and subsequent economic calculation.
- The word 'energy', used later in the report, will signify electricity unless stated otherwise.

4 Literature review

This chapter highlights the current state of research on the topic. It synthesises and evaluates the existing material and information.

Denmark is a global leader in PtX development. It is targeting 4–6 GW of electrolysis capacity by 2030 to support the 70 % CO_2 reduction target and to achieve climate neutrality by 2045 [16]. The 2020 Climate Programme positions Denmark as a hub for green fuel and hydrogen exports, taking advantage of its excellent offshore wind resources and the fact that it has over 30 PtX projects underway. This strategy involves integrating PtX with energy islands, such as BEI, to convert surplus renewable energy into fuels for hard-to-abate sectors, such as aviation and shipping. Energy islands are a key part of Denmark's PtX strategy, combining large offshore wind turbines with electrolysis. This helps to deal with the problems of including different types of renewable energy and producing large amounts of green fuel [18].

Singlitico et al. (2020) are focusing on the feasibility and optimal placement of PtX facilities in conjunction with BEI's offshore wind power expansion [19]. Two years later, the Danish Transmission System Operator (TSO) Energinet released a business case for BEI's Energy Infrastructure. The report's findings suggest that connecting BEI to both Denmark and Germany would offer significantly greater economic and climate benefits than connecting only to Denmark. This would result in a nine-figure net benefit and reduced CO_2 emissions (40 000 tonnes in Denmark and 3,1 million tonnes in the EU by 2040). Energinet advises using HVDC technology for a hybrid interconnection between the two countries, with either a synchronised or phased completion timeline by 2029–2030. The BEI project is advancing the green transition by generating large volumes of renewable energy for use in electrification and the production of green fuels. It could also be connected to other EU countries in the future. Risks have been identified, including ambitious time schedules, undeveloped technical standards for HVDC, uncertain market conditions, and regulatory challenges. The economic outcome is subject to future changes in CO_2 prices, electricity demand for PtX,

renewable expansion, and broader EU climate ambitions [20].

The National Center for Green Energy of Denmark (2023) stated that Bornholm is presented as an optimal location for the experimental evaluation and demonstration of energy island technologies. This is in support of research and innovation in the areas of offshore wind, PtX, battery storage, and digital solutions for grids. The report outlines scenarios for testing physical and digital technologies, focusing on grid integration, system optimisation, and control strategies. Bornholm's current energy system and infrastructure offer a pragmatic case study for the development of future energy island configurations, harbouring the potential for distributed and large-scale demonstration projects [21].

Two more relevant reports were released in that same year. The first examined the potential impact of a PtX facility on the local community and surrounding area of Bornholm. As well as having the capacity to generate employment opportunities and stimulate economic growth in Bornholm, PtX was found to enhance energy independence through sector coupling [22].

Secondly, Hedetoft et al. provide quantitative projections of the local impacts of the BEI project using QBIS (workforce modelling) and SAM-KLINE (input-output analysis) across three scenarios. The direct effects include 600 person-years of construction for the wind farm (60 of which would be local roles), 1 775 person-years for the onshore station (80 of which would be local roles), and up to 3 300 person-years for a hypothetical 3 GW PtX facility (246 of which would be local roles). These would generate 190–900 permanent jobs and lead to a population increase of 460–1 059 by 2030. Under full development, indirect effects would generate an annual Gross Value Added of 793 million Danish krones. However, prerequisites such as expanding vocational training and providing temporary housing for 5 298 imported workers would need to be met to avoid disruption to tourism [23].

Lüth and Keles (2024) provided a comprehensive literature-based survey on OEHs, focusing on their risks, strategies, and benefits. They found out that OEHs

are emerging as a vital component in the decarbonisation process of energy systems. These hubs facilitate the integration of offshore wind farms, interconnections, and conversion technologies such as electrolysis, along with energy storage systems, thereby enabling the transition towards a low-carbon energy future. Although offshore wind power has already reached a state of technological maturity, they state that other components of the wind energy production process, such as offshore electrolysis and large-scale storage, remain technologically risky and economically challenging due to high costs, unclear market frameworks, and complex ownership structures. OEHs also face ecological trade-offs, such as reducing emissions while managing the impact on biodiversity and resources, as well as societal issues relating to public acceptance and limited local involvement. Coordinating cross-border planning is of critical importance, yet this process is complicated by geopolitical and regulatory factors. To achieve strategic success, a multifaceted approach is required, including balanced integration, social equity, and the advancement of technology, policy, and market design through further research [24].

Hansen et al. (2024) analysed how sustainable transitions unfold locally using BEI as a case, highlighting both economic and social opportunities alongside demographic challenges. The study stresses that success depends on managing diverse stakeholders, ensuring strong institutional and policy support, and adopting inclusive, place-based approaches in a peripheral rural community [25].

Mörchen et al. (2025) compared how different interconnection configurations through BEI affect Denmark, Germany, Sweden and Poland. A fundamental asymmetry in project benefits has been shown by their analysis: while reductions in system costs, fuel consumption, and CO_2 emissions are achieved by Germany, Poland, and Sweden across all scenarios, increased costs and emissions are experienced by Denmark regardless of the configuration. It is notable, that Scenario 3 — connecting all four countries — achieves the lowest system costs and emissions, but that leaves Denmark economically disadvantaged. The authors conclude that realising the cross-border potential of BEI re-

quires addressing this uneven distribution of benefits through fair cost-sharing and benefit-redistribution mechanisms, emphasising that technical optimisation alone cannot resolve the underlying economic disparities between participating nations [26].

Manniche and Hansen (2025) examined how Bornholm could develop as an environment for testing and demonstrating green energy technologies, linked to the planned BEI project. Drawing on 15 years of experience with test islands, the authors propose three scenarios: a technology testing environment, a centre for developing sustainable energy systems, and a living laboratory for sustainability transitions on a community scale. Bornholm has many assets that could facilitate innovation. These include an established collaboration between the Technical University of Denmark (DTU) and Bornholms Energi og Forsyning. The island also has supportive local policies and university networks. However, as a small rural municipality that is experiencing demographic decline and has limited institutional capacity, making the most of these opportunities will require coordinated municipal planning, public–private partnerships, the involvement of local businesses and support from national policy. The paper emphasises that creating a meaningful green path and generating local economic benefits depends on integrating technological, social, and institutional innovations across all sectors of the island economy [27].

Taken together, these studies show that while BEI has been examined from the perspectives of governance, local socio-economic impacts, interconnector business cases, and high-level system benefits, there is still a lack of detailed, bottom-up techno-economic scenario analyses comparing alternative development pathways of Bornholm as an OEH (with and without PtX), which is the gap this report seeks to address.

5 Methodology

The following chapter outlines the methods and research tools employed to address the research questions. The methodological framework begins with a presentation of the case study, prompted by the problem statement, which identifies Bornholm as the optimal example for analysis. Based on the literature review that precedes it, the chapter outlines a semi-structured expert interview that was conducted to deepen the understanding of key technological and economic aspects, and to define a concrete analytical focus. Furthermore, scenarios were introduced to provide a more comprehensive overview of the evaluation process for the project. The models themselves, which are developed based on the scenarios created, utilise EnergyPro as the modelling software best suited to the project.

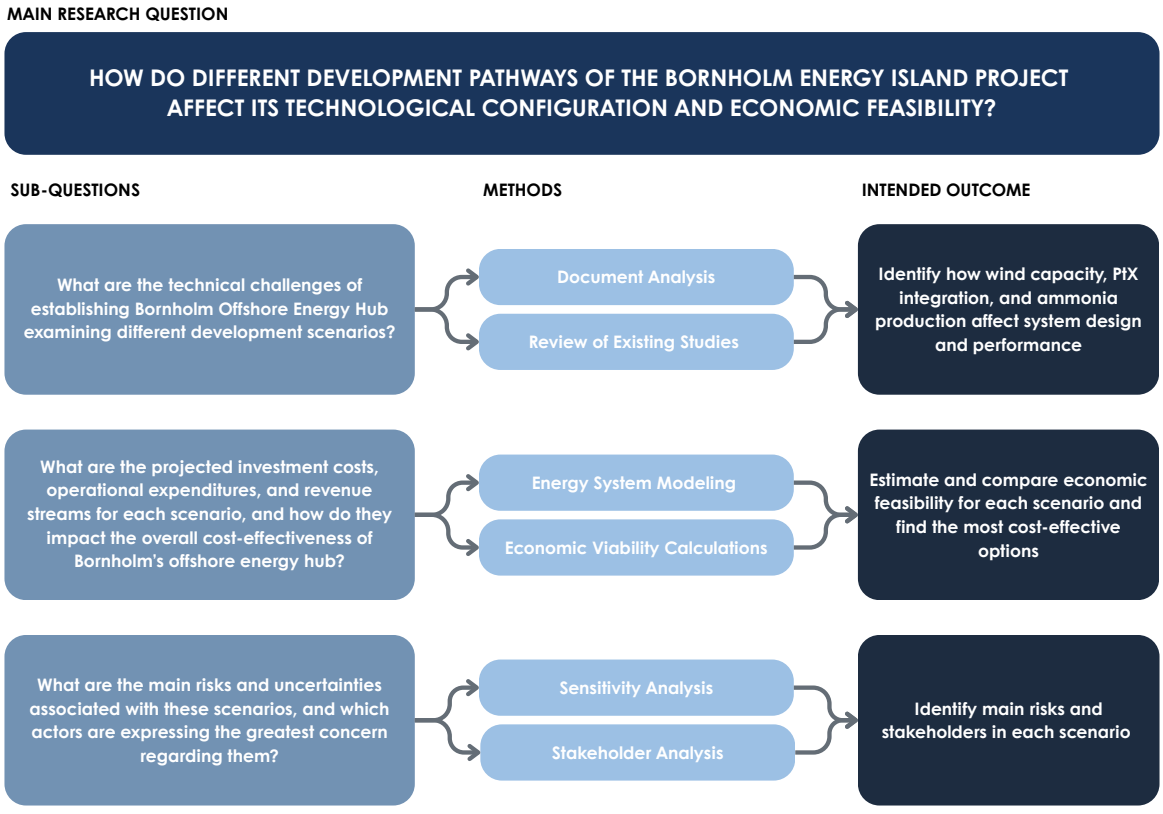


Figure 5: Research question (source: own work)

5.1 Case study - Bornholm Energy Island

Bornholm is a Danish island located in the Baltic Sea, approximately 150 km east of Zealand, 50 km south of Sweden and 100 km north-east of Germany (see Figure 6). Its area is over 580 km² with a population of almost 4 000 people [28].

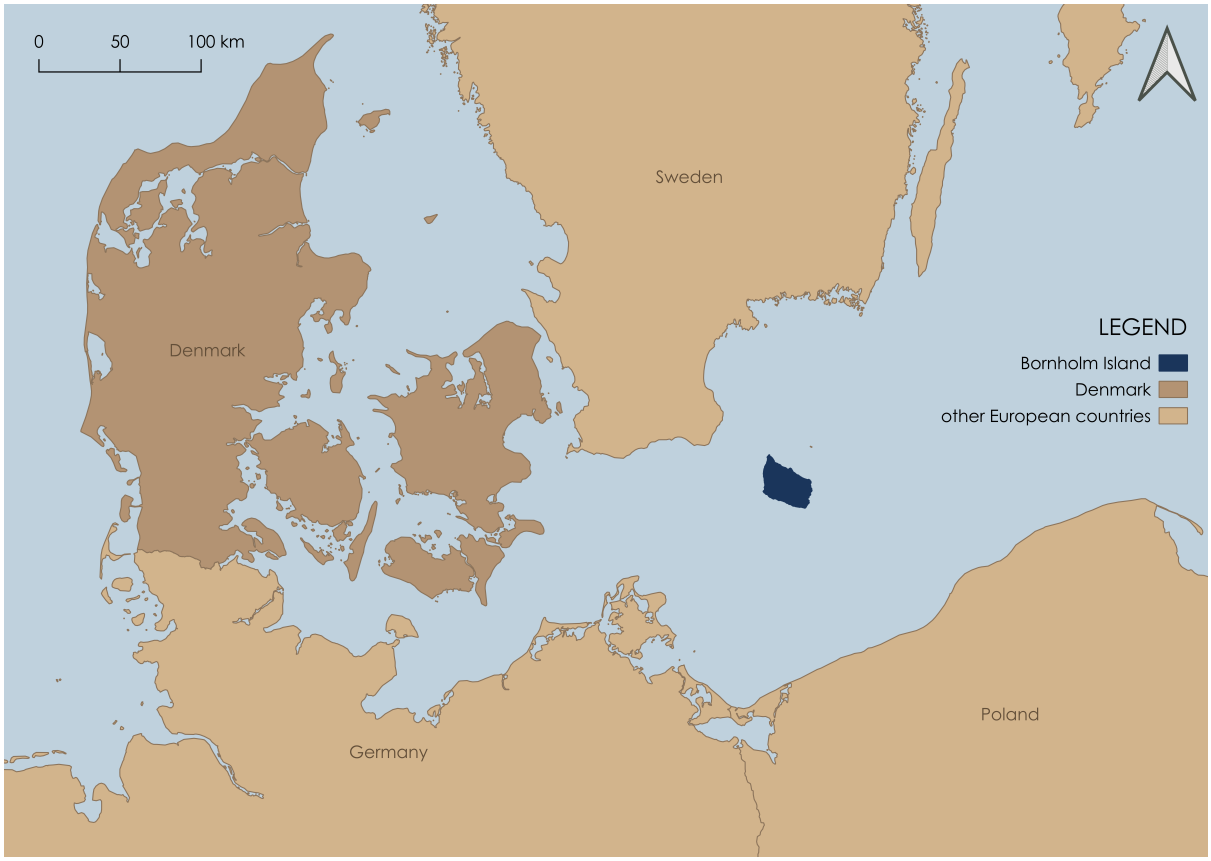


Figure 6: Bornholm Island, situated off the coast of Denmark and other European countries (source: own work)

The BEI project is one of Denmark's biggest initiatives in the transition to a fully renewable energy system. The island is set to become a central hub in the Baltic Sea, combining offshore wind development with advanced grid infrastructure and new energy applications. The project is currently in the early stages of construction, with the latest development (as of October 2025) being the contract between NKT and Energinet to install a HVDC cable connecting Bornholm to Zealand [29].

The underwater cable connections are one of the already-approved and set-in-motion parts of the Energy Island project, providing a link to Zealand, as

well as mainland Germany (developed together with the German TSO 50Hertz). The Bornholm - Zealand cable will have a capacity of 1,2 GW, while the Bornholm - Germany connection is planned to have 2 GW [17]. Those connections will establish a cross-border energy system and create the foundation for the expansion of offshore wind capacity in the Baltic Sea. Official documents have also noted the potential for future connections to Sweden and/or Poland [12].

Another key component of the project are offshore wind farms which are planned to have a capacity of either 3,2 GW or 3,8 GW. The baseline political agreement was for 3 GW but the higher figures reflect the ones from the official documents. The capacity is either 3,2 GW or 3,8 GW depending on several external factors, such as the shape of the seabed, shipping lanes, technicalities, etc.

For the purposes of this report, the capacity of 3,2 GW will be taken as the baseline for all calculations and simulations. This is because it reflects the confirmed underwater cable connections to the countries under consideration.

The three planned offshore wind farms are Bornholm I Nord, Bornholm I South and Bornholm II (see Figure 7). The total area of the site is 651 km², and it is located at a minimum distance of 15 km from the coast of the island [17]. This is regarded as the baseline for future scenarios. Moreover, official planning documents and analyses available on the Energistyrelsen also consider the future possibility of connecting the island hub to additional wind energy sources from the sea [12]. These documents do not specify whether the additional wind farms would be the ones already existing in the Baltic Sea, their infrastructure connected to the one on Bornholm, or if they would be, similarly, built specifically for this project.

The project's timeline assumes that the OEH, with the wind farms and cable connections, will be operational by 2030, which is an ambitious goal. The rest of the infrastructure and possible expansions will come in later years.

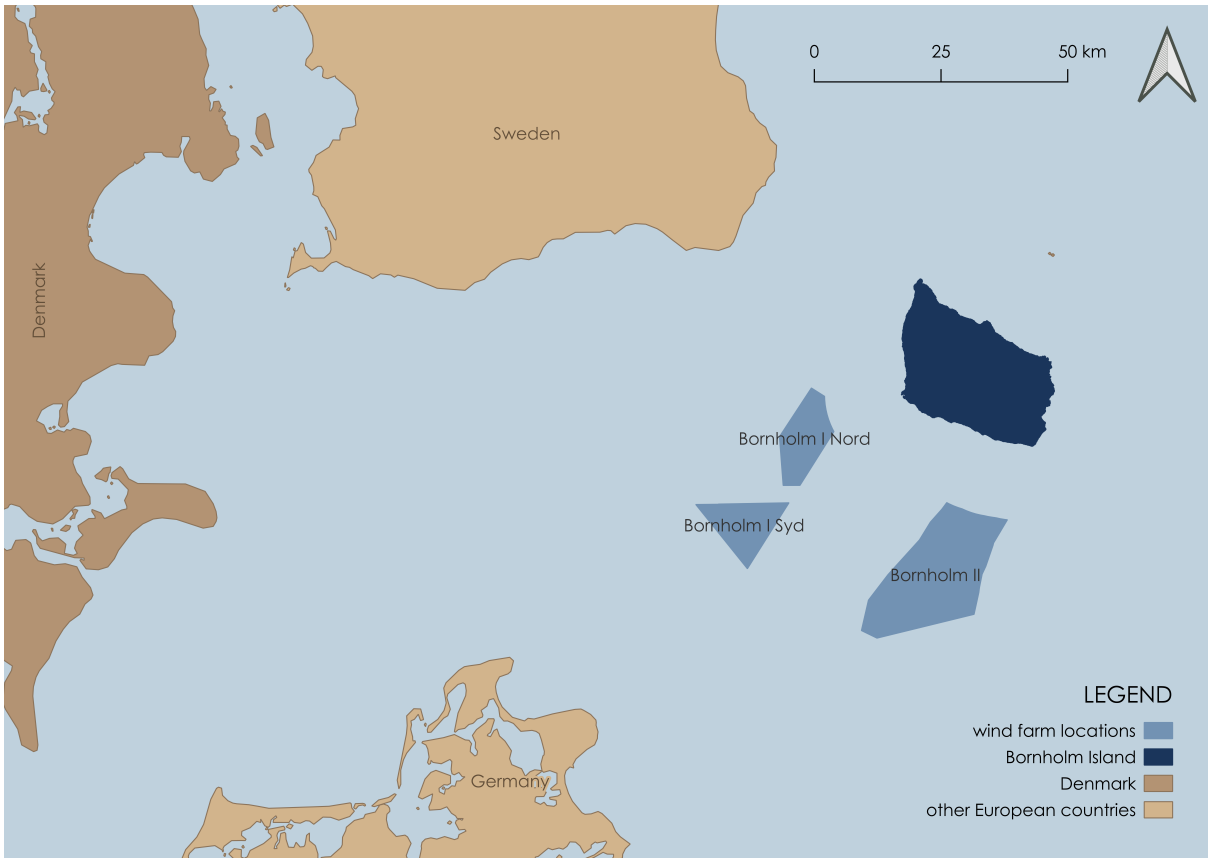


Figure 7: Location of the planned wind farms near Bornholm (source: own work)

In addition to transmitting renewable energy to Denmark and Germany, BEI is expected to have PtX facilities. These installations would convert extra wind power into green hydrogen or other e-fuels. Such conversion supports grid balancing, enables energy storage, and creates feedstock for sectors that are difficult to electrify, such as heavy industry, shipping, and aviation.

Additionally, plans include the development of a fueling station for ships, where boats can get sustainable fuel coming from PtX production. Since around 60 000 ships pass the Baltic Sea annually [30], this could position Bornholm as a key player in the decarbonisation of international shipping routes.

There are many other potential expansions or additions to the BEI, including using the green energy for district heating, biogas implementation, creating an energy-intensive industry sector, etc. Overall, the project has a big potential and it offers much more than just offshore wind energy. It represents an innovative approach to integrating renewable power, enabling cross-border cooperation, and pioneering new applications such as green marine fuels.

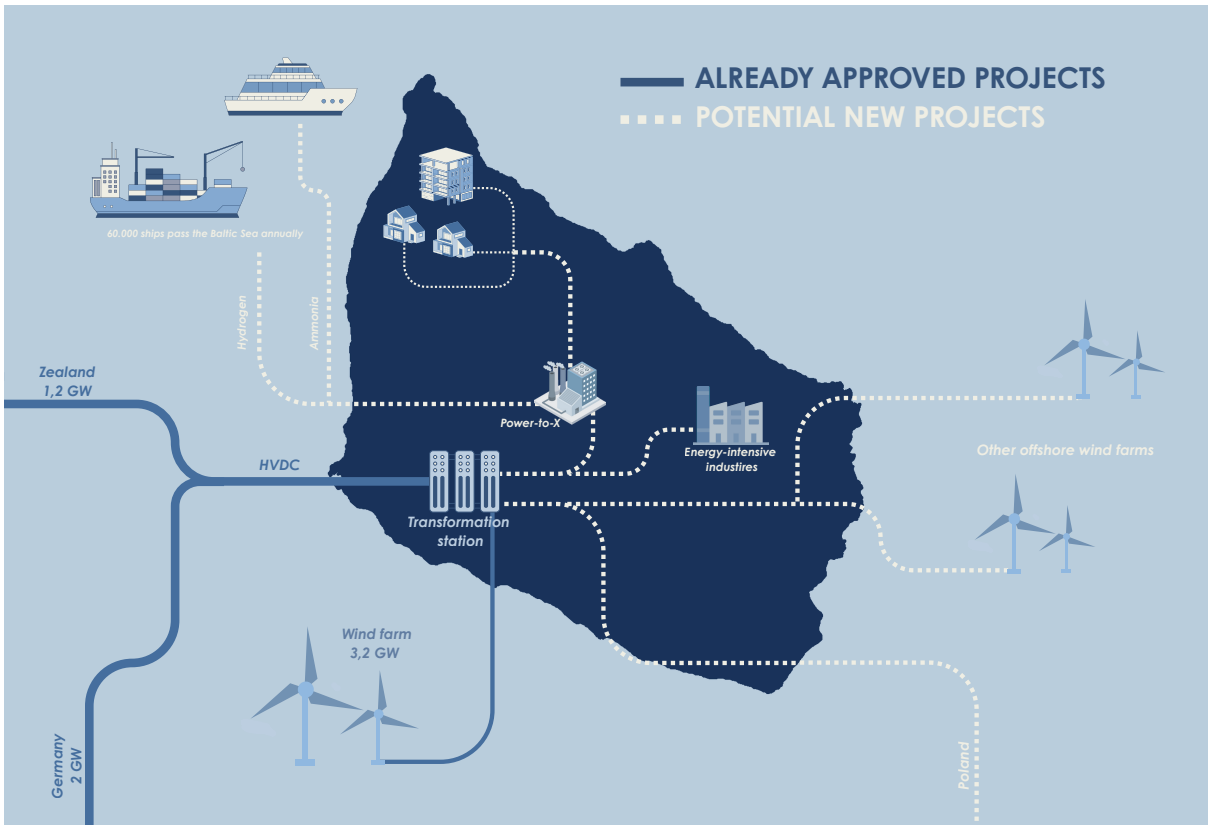


Figure 8: The concept of Bornholm Energy Island (source: own work inspired by [31])

5.2 Interview and conference

The preliminary expert interview was conducted in the early stages of the project. The primary objective of this initiative was to refine the methodology and to acquire novel insights into the function of energy hubs. The observations presented into the subject matter were made possible thanks to an interviewee with experiential knowledge of working with these hubs. The interview was concluded with Iva Ridjan Skov, Associate Professor at Aalborg University (AAU) Copenhagen, involved in, among others, the Nordic Hub project (full transcript in Appendix - 10.1).

It was through this process that a significant amount of valuable input was gained. Most importantly, it was evident that PtX had already been considered at the project's early stage. In addition, it was confirmed that PtX aligns well with Bornholm's strategic objective of becoming a fuel distribution hub. It was established that both methanol and ammonia could be considered as potential fuels. However, a significant challenge for producing the first one is the require-

ment for CO_2. The potential for hydrogen exploitation was a subject of discussion, with the introduction of hydrogen ferries to the region being a possibility.

While working on the report, the project group had the opportunity to attend the Energy Island Summit 2025, which took place at the beginning of December on Bornholm. This event brought together political decision-makers, regulators, industry CEOs, and leading researchers to highlight the importance of Europe's energy independence, competitiveness, and collaboration in the energy transition [32]. The primary objective was to develop a deeper understanding of projects relating to OEHs, and to gain insight into the major technological and economic challenges they present.

During the Energy Island Summit 2025, European Commissioner for Energy and Housing Dan Jørgensen stated that the project is crucial from international, national and local perspectives. He also emphasised that it needs to be implemented soon. This is especially important in light of the current geopolitical situation in Europe. The aim is to strengthen countries' energy security and resilience.

Political representatives from Denmark and Germany — including Parliamentary State Secretary of Germany Stefan Rouenhoff, Member of the Danish Parliament Lea Wermelin, and Deputy Permanent Secretary at the Danish Ministry for Climate, Energy and Utilities Christian Stenberg - reaffirmed that the Danish–German partnership, with BEI as a shared lighthouse, is a strategic pillar for both countries and Jørgensen's broader European approach.

Industry CEOs from 50Hertz, Energinet, Siemens Energy, and NKT Cable adopted a remarkably unified approach. Their message was clear: Europe is moving from the national to the regional level in terms of the development of its energy system. This shift is essential to unlocking the full potential of offshore wind and using renewable energy efficiently. It is also crucial for building the redundancy that the European grids urgently need — and it can only succeed through coordination, certainty, and collaboration.

Danish wind power pioneer Henrik Stiesdal presented a clear pathway toward a truly energy-independent Europe. He underlined why energy islands are a strategic and necessary part of this future: they combine energy production, energy transmission, and energy security in one integrated asset.

Furthermore, the project must not only proceed, but also succeed, as its failure could undermine the development of all other planned energy island projects — a point highlighted by Nicolaos A. Cutululis, Professor at DTU.

In summary, the Energy Island Summit 2025 reinforced BEI's pivotal role as a lighthouse project for Danish-German collaboration and European energy independence. The summit united political, industrial, and academic voices in calling for swift, coordinated regional implementation amid geopolitical challenges. There was a clear consensus that its success is essential for unlocking the potential of offshore wind and grid redundancy, as well as paving the way for all future energy island projects across Europe.

5.3 EnergyPro

EnergyPro is a professional energy system analysis software developed by the Danish company EMD International A/S. It is frequently used in the energy sector, consulting, and research for the technical and economic assessment of energy projects.

The software allows for detailed simulation of electricity, heating, cooling, and fuel production within a single modelling environment, combining technical performance data with economic optimisation to evaluate both operational strategies and investment feasibility. The software operates on hourly or sub-hourly time-series data, which allows it to capture the variability of renewable energy generation, market prices, and fuel costs, and to determine the optimal system operation under given conditions.

In the context of this project, EnergyPRO version 5.0 is used to model and compare alternative development pathways for the BEI. Each model reflects a different scenario — ranging from the approved baseline to systems includ-

ing expanded offshore wind power and PtX technologies, as well as ammonia production. Through the use of EnergyPRO, the project aims to assess the technical performance, energy balance, and economic viability of each configuration, ultimately providing a comprehensive evaluation of how different system designs could contribute to the sustainable development of the BEI.

5.4 WindPro

WindPro is a professional software designed for planning, design, and analysis of wind energy projects. Developed and distributed by the same company that made EnergyPro - EMD International A/S - it is used for similar purposes: consulting, research, wind-farm project assessment and optimisation.

This software enables for a more in-depth analysis of wind-based energy projects by integrating extra tools that help examine both the technical and environmental aspects of wind-power development. Apart from basic tools to assess productivity or the economic aspect, WindPro also allows for: integrating wind speed data, noise and glare analysis, wake losses calculation, visualisation of layouts based on Geographic Information System (GIS), and more.

In the context of this project, WindPro version 4.2 is used to assess potential wind-farm configurations specified for the BEI. By simulating multiple turbine arrangements and operational conditions, the software will allow to compare the scenarios with each other; their efficiency, feasability, cost, etc. This analysis complements the broader system-level modelling performed in EnergyPRO, ensuring that all the scenarios and their implications are comprehensively understood.

5.5 Scenarios development

Initiatives that are designed to introduce innovation, new technology, or solutions to the market can have a considerable impact on the energy sector, both with regard to emissions, energy consumption, and the overall political and technological situation. It is important to correctly identify possible im-

provements and implement changes in line with anticipated outcomes. Therefore, to properly understand the role of the OEH on Bornholm, an analysis of potential projects to be implemented on the island (referred to in this study as 'scenarios') will be conducted. In the process of establishing particular scenarios, it is essential to consider the fundamental driving forces, select the most critical ones, assess the range of scenarios, and develop consistent logic to ensure the scenarios remain coherent.

- **Driving forces**

  The primary objective of the present study is to analyse technological and economic factors, with the environmental aspect being the main objective that guides the identification of solutions. The political, social, and legal aspects are of secondary importance and will not be analysed in the scenarios.

- **Scope**

  In light of prior analysis regarding the current situation in Europe and the pursuit of the most environmentally friendly solutions, the scope has been narrowed to the utilisation of green energy and fuels.

- **Space**

  The base for the research is part of the project that has already been approved. The target size and type of other technologies that can be used for the OEH are currently unknown. The fundamental question guiding our research is whether it is worthwhile to pursue the expansion of the project and, if so, in which direction.

- **Logic**

  The four scenarios presented propose solutions that reflect the next steps to be taken on the island and are aligned with our assumptions. It is important to maintain consistency so that the specific logistics and details make sense in terms of scope, space, and the correlations between the scenarios themselves [33].

A wind farm with an installed nominal capacity of 3,2 GW is currently planned to be included in the project. This will form the basis of the project, which will be referred to as the 'baseline' later in the report. To ensure consistent logic, the potential expansion of the project will comprise plans for additional wind turbines around the island and PtX, which involves creating green fuel on site to meet the demand of ships.

As an island, Bornholm offers ideal conditions for large-scale wind farm that can be connected to the island and other countries via the OEH. The favourable wind conditions and a relatively high degree of freedom in choosing locations make offshore wind turbines a more attractive option than other RES.

Plans are in place to establish a green bunkering station at Bornholm. Utilising on-site renewable energy production, the introduction of PtX and green fuel production is the next step towards an energy port focused on green transformation and the bunkering of green fuels [34]. In order to focus on the use of renewable energy, taking into account its availability and desire to remain on a carbon-neutral path, ammonia was chosen as an environmentally friendly fuel.

The final version of the 2-dimensional Scenario Space is presented below:

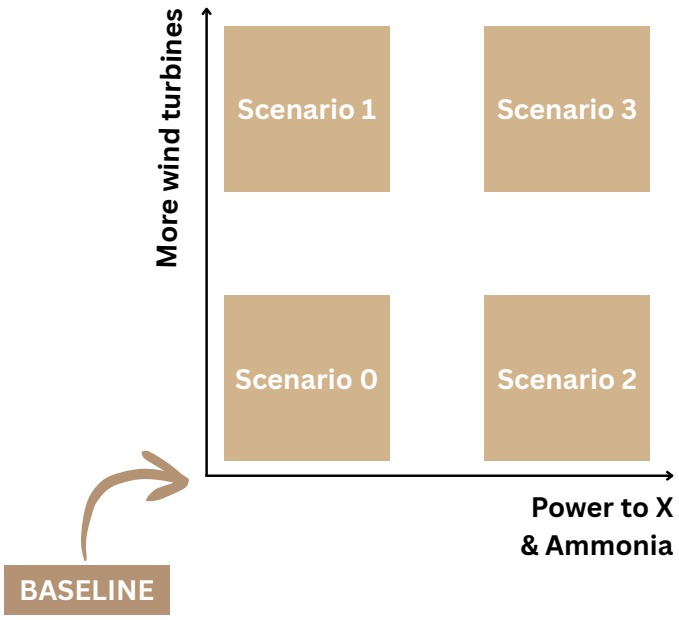


Figure 9: Scenario space (source: own work inspired by [33])

6 Technical and economic assumptions

6.1 Technical analysis

This project considers an offshore wind farm and PtX technologies. The following section provides a detailed description of the technical assumptions and methodological framework applied in this study.

6.1.1 Offshore wind farm

The energy hub's production originates from the wind farm, initially estimated at 3,2 GW and subsequently scaled up to 4,8 GW in later scenarios. The wind turbine model is expected to have an output of approximately 15 MW. It is proposed that the technical parameters of similar wind turbines will be used for the simulation. In consideration of the power curve, the following curve is potentially comparable:

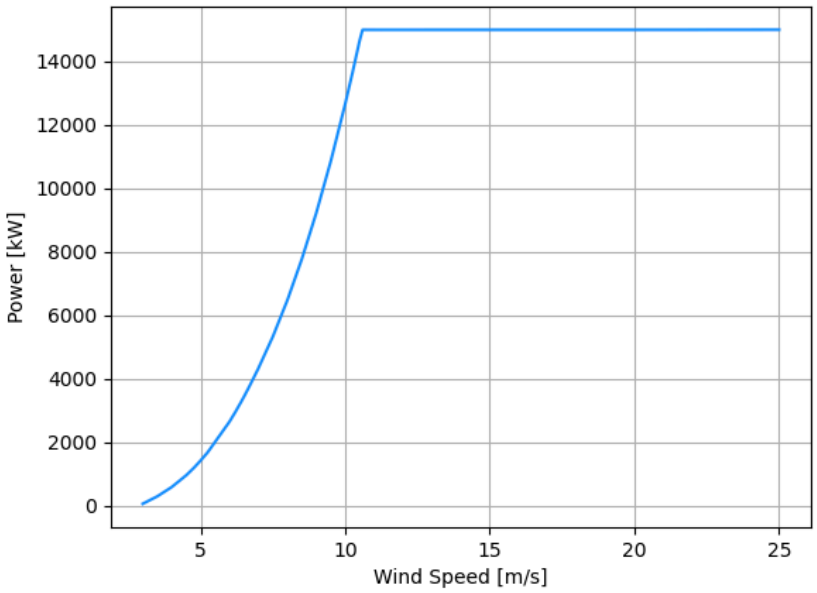


Figure 10: Power Curve of the wind turbine IEA 15 MW 240 RWT (source: [35])

6.1.2 Cable connections

The electricity generated by the offshore wind farms located off the island's coast will be centralised in this hub, converted, and transported via 525 kV

HVDC submarine and land cable systems to substations located in Zealand (Denmark) and Mecklenburg-Western Pomerania (Germany). The Company NKT has been selected as the supplier for the 525 kV HVDC system [36].

6.1.3 Hydrogen

The first step of the study was to select the most appropriate electrolysis technology between Alkaline Electrolysis and Proton Exchange Membrane Electrolytic Cells (PEMEC).

Alkaline electrolysers represent the most mature and widely used technology. They offer several advantages, such as technological maturity, wide market availability, lower investment costs, and a long operational lifetime of approximately 100 000 hours. However, their main limitation lies in their reduced suitability for operation under fluctuating power inputs. When frequent start-up and shut-down cycles occur, such as in systems powered by wind turbines, the performance and durability of alkaline electrolysers tend to decline [37].

In contrast, PEM electrolysers are a more recent technology characterised by a fast dynamic response, making them particularly suitable for integration with RES. They also provide higher hydrogen purity, reaching up to 99,99 %. However, their main drawbacks are the higher capital cost and a shorter lifetime, around 80 000 hours between membrane replacements. Nevertheless, the PEMEC technology was selected in this project to ensure optimal coupling with the energy island [37].

The fundamental operating principle of the PEMEC is presented in Appendix 10.2.1. In general, PEMEC electrolysers achieve an efficiency of about 40 %, which may vary within ±5 % depending on the input power fluctuations [37]. For the sake of simplification, an average efficiency of 40 % is considered in this study. To maintain high efficiency, multiple electrolysers of appropriate size can be operated in parallel. This configuration allows the system to start up or shut down individual units to remain within the optimal efficiency range. Moreover, operating several electrolysers enables better thermal management, keeping

them within their ideal temperature window and avoiding efficiency degradation. For simplicity, the hydrogen outlet pressure is assumed to be 10 bar, although in practice it can range between 5 and 30 bar, with 10 bar being the most common operating condition [38]. In addition, it is assumed that a continuous supply of deionised water is available to the electrolyser, and that water purification constraints are not considered within the scope of this study.

6.1.4 PtX

Scenarios 2 and 3 both take into account PtX technologies. According to the interview, presented fully in the section 10.1, the expense associated with the production of carbon dioxide for the purpose of methanation or methanol production is not economically viable in the case of Bornholm. The project's primary focus is on ammonia production, utilising hydrogen and nitrogen as key components.

Around 175 million tonnes of ammonia are produced each year worldwide. Forecasts indicate that demand will increase in the coming years. Unfortunately, the Haber-Bosch process, which relies on fossil fuels, is responsible for up to 96 % of the production [39]. This process consequently results in carbon emissions. In contrast, novel designs for the Haber-Bosch process are being developed for integration with RES. The use of ammonia as a hydrogen carrier would facilitate its transportation, with no CO_2 being emitted when the hydrogen is released. Furthermore, ammonia could be used as a fertiliser in agriculture and even as fuel for ships. The development of ammonia-fuelled ships has the potential to decarbonise the global shipping industry, which accounts for more than 80 % of worldwide shipping [40].

The Green Haber-Bosch process is a modified version of the Haber-Bosch process that has been adapted for use with RES. This results in a process that achieves different parameters to those of fossil fuels, primarily due to the use of lower temperatures, smaller-scale infrastructure and lower pressure values. The process under consideration in this study will require hydrogen, nitrogen, water, electricity, and a certain amount of cooling infrastructure in order to

facilitate the production of ammonia (see Figure 11). In summary, the production of 10,1 kg of 99,95 % pure ammonia is achieved from a mixture of 1,8 kg of hydrogen and 8,5 kg of nitrogen, following the chemical reaction:

$$3\,H_2 + N_2 \longleftrightarrow 2\,NH_3 + 92{,}44\,kJ$$

The discrepancy in mass can be attributed to recycling loops. The production of ammonia is favorised at elevated temperatures and pressures (450 °C and 200 bar). However, the use of a catalyst allows for a reduction of these parameters to 200 °C and 80 bar, thus rendering the process more suitable for the utilisation of non-fossil fuels. The electrical consumption of the process is directly linked to the utilisation of compressors and pumps, with the consumption amounting to 0,28 kWh per kg of NH_3 produced. The thermal energy consumption is only 0,001 kWh of cold per kg of NH_3. Heat recirculation loops allow the exothermal property of the reaction to be used to heat the inlet gases, while cooling the outlet gas.[41]

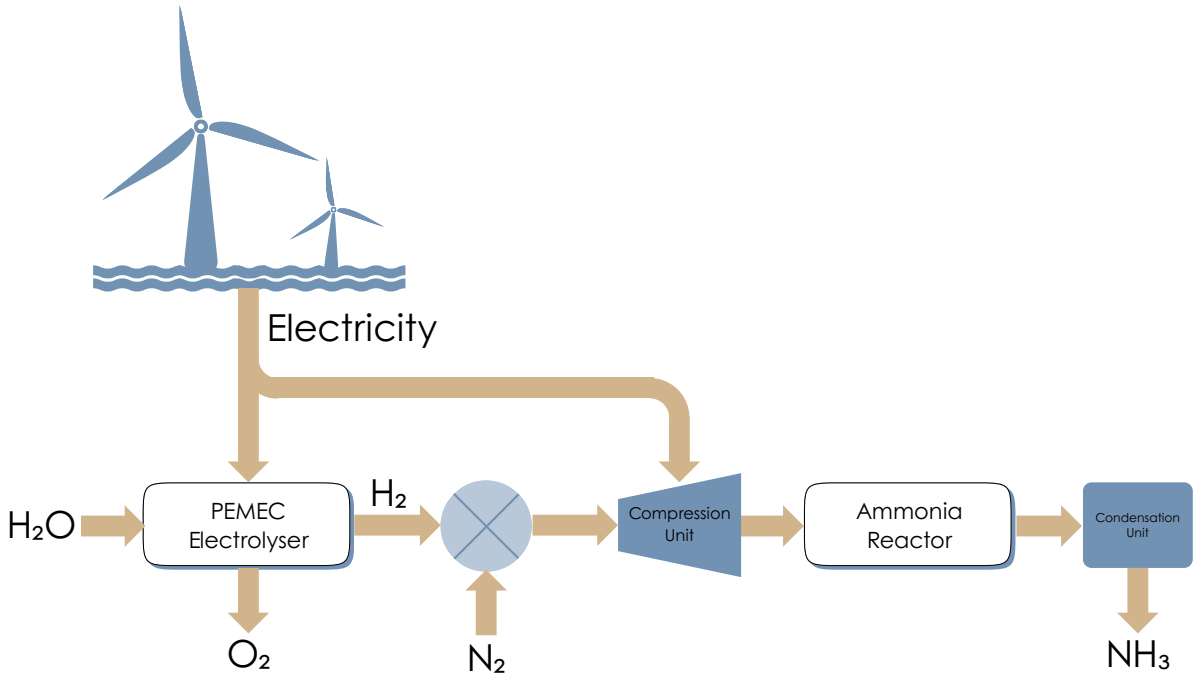


Figure 11: Power to Ammonia - simplified schematic (source: own work)

6.2 EnergyPro models

The EnergyPro model comprises three primary sites: As demonstrated in Figure 12, the OEH on Bornholm is connected to Denmark and to Germany. The electricity is subsequently sold in the day-ahead market, with separate prices applied for the two countries. The historical prices from 2024 are used in the simulations for all the scenarios, as they represent the newest full year data that could be used for the project. The prices for Denmark and for Germany were published on the website Energy Prices [42].

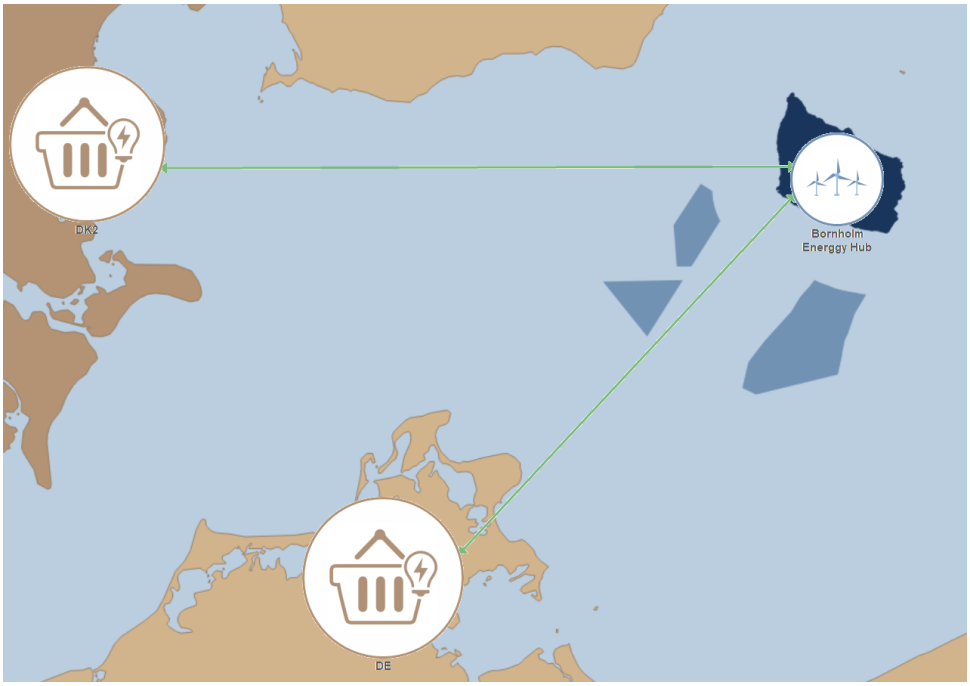


Figure 12: EnergyPro - site overview (source: screenshot from EnergyPro)

6.2.1 Scenario 0 - Baseline

In this scenario, the focus is exclusively on projects that have already been approved. This involves the construction of an offshore wind farm with a capacity of 3,2 GW near Bornholm [17], as well as the cable connection to Denmark and Germany. The electricity produced would be distributed through the cables to the countries. The scenario forms the baseline, which will be used to compare the current plan with other possible outcomes.

The model in EnergyPro was developed on the basis of these assumptions, in combination with other available data concerning the technologies employed. For the analysis of wind farm data, WindPro was utilised. This program offered the capacity to retrieve data, including local wind statistics, which were then used to provide precise results regarding wind farm energy production. The quantity of wind turbines, the area designated for their installation, and the proposed distribution thereof were all derived from analyses that had previously been conducted and verified. The wind power curve data for the wind turbines was established using data illustrated in Figure 10. It is envisioned that, with partial load and transmission to other sites permitted, this will form the basis of the analysis and will be the sole energy producer.

It is asserted that the separate electricity markets are established and will not be subject to alteration in the context of any other scenario. Operation and maintenance (O&M) costs are incorporated into this version for the purpose of evaluating wind turbine performance. The short-run marginal cost of an O&M wind turbine is determined by the following factors:

- the fixed O&M cost of €60 000 per MW per year and the variable cost of €3,5 per MWh per year was assumed [43];
- the CO_2 emissions associated with O&M, which are calculated at 7 gCO_2 per kWh [44];
- the CO_2 price, which is set at €75 per tonne; value assumed based on changes in 2025 [45].

It is generally assumed that the full utilisation of energy is possible, with the possibility of energy being both exported, imported, and used for ammonia production. To avoid a large-scale import and to get as close as possible to the real-life circumstances, it was assumed that grid losses of 3 % apply when transmitting energy through a submarine cable (the figure has been calculated on the basis of 390 km of cable length in total and a total capacity of 3,2 GW [46]).

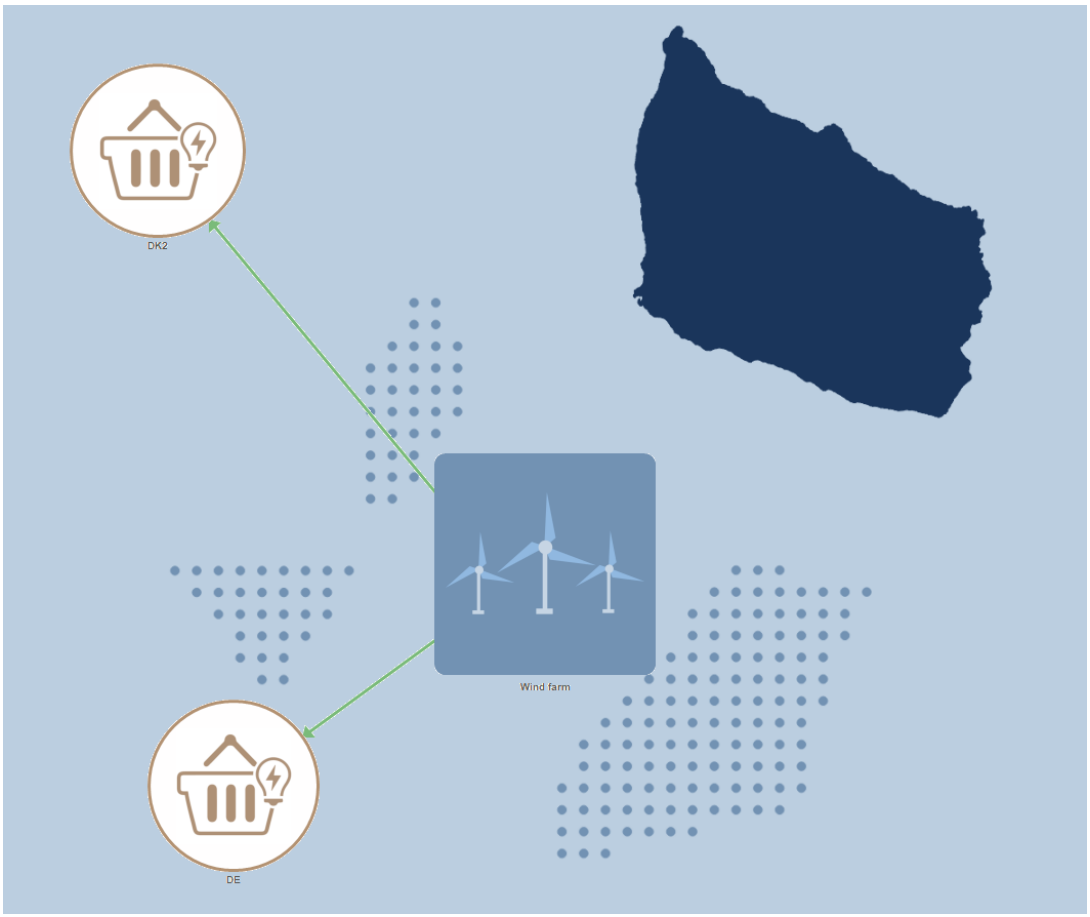


Figure 13: EnergyPro - site overview, scenarios 0 and 1 (source: screenshot from EnergyPro)

6.2.2 Scenario 1

In the Scenario 1, the initial project is expanded to include a greater number of offshore wind turbines. The analysis will address the construction and installation of the equipment, as well as its subsequent operation. It is predicted that the supplementary space on the water around the island can be utilised, in accordance with the analyses that have already been carried out for the initial phase of the project.

This scenario involves expansion, meaning the construction of 50 % more wind turbines than in the baseline scenario. An additional 1,6 GW of wind power was modelled in WindPro, creating a 4,8 GW wind farm. The data for O&M and CO_2 costs remain unchanged, as do previous assumptions.

To maintain flexibility, it is assumed that additional cables or cables with

greater electric power transmission capacity will be laid to facilitate the transmission of greater energy.

6.2.3 Scenario 2

This scenario involves using approved energy-producing devices to sell energy to other countries and produce green fuel. The fuel production process involves generating hydrogen via electrolysis and synthesising ammonia with supplied nitrogen. The entire process is powered by on-site wind energy.

Additional modules had to be created within EnergyPro. Two new fuels were defined: water and nitrogen. Also, two new units representing an electrolyser and ammonia synthesis were added, for which capacity was the main parameter specifying their operation. The electrolyser has a capacity of 647,1 MW and the ammonia synthesis has the ability to produce 10 tonnes of ammonia per hour given it's capacity of 86,3 MW. By providing the programme with the specific investment cost in € per MW and the expected lifetime of the unit, the programme was able to select the unit size. The further capacity computation and optimisation are explained in section 7.2.4 considering investment cost based on data sheets for renewable fuels [47]. In the electrolyser, a water input of about 167 kg/h [48] is needed to produce 17 kg/h [47] of hydrogen at 10 bar. Calculations were made based on the higher heating values of hydrogen (141,7 MJ/kg) and ammonia (22,5 MJ/kg) [49].

The following additional prices were used:

- nitrogen purchase: €0,11 per kg [50];
- fixed O&M costs for electrolyser: 2% of specific investment per year [48];
- fixed and variable O&M costs for ammonia synthesis: €45 per MW per year and €0,02 per MWh, respectively [48];
- ammonia sale: €0,75 per kg [51].

In addition to the previous assumptions, it was considered that access to water was unlimited and free of cost for use in the electrolyser. It was also assumed

that nitrogen would be available as a fuel source, rather than investing in a unit for separating air.

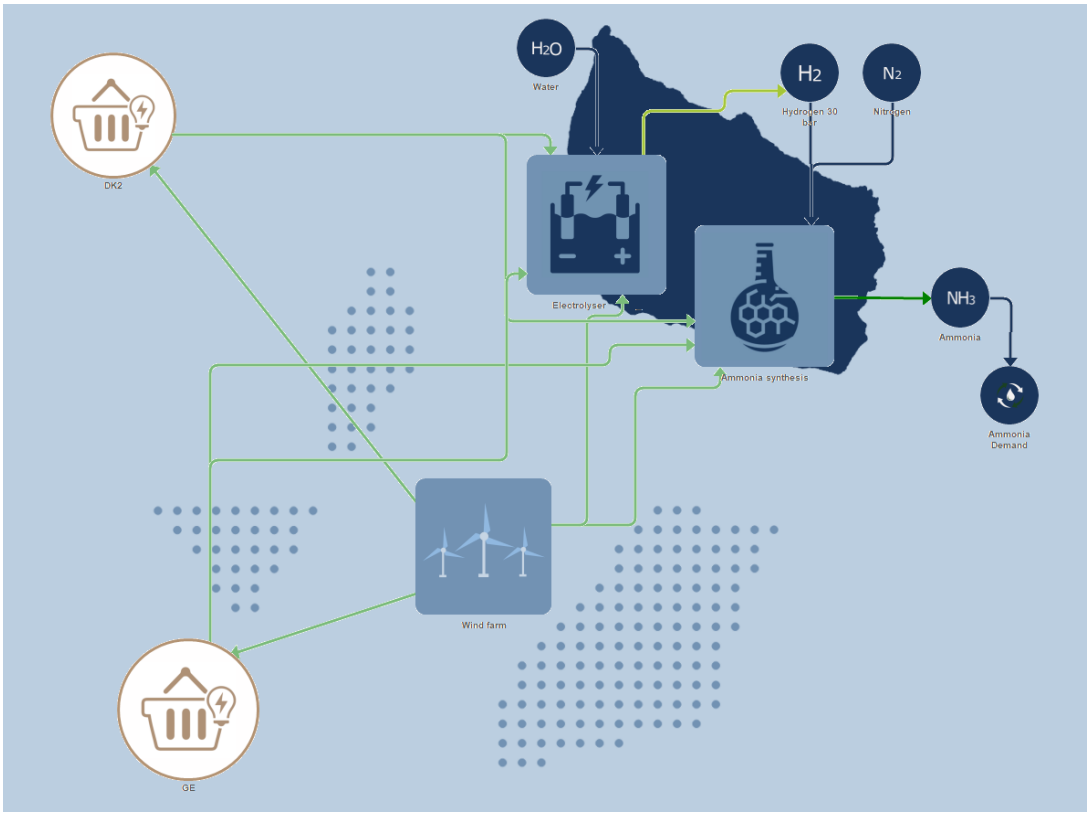


Figure 14: EnergyPro - site overview, scenarios 2 and 3 (source: screenshot from EnergyPro)

6.2.4 Scenario 3

The final scenario involves expanding the project to include additional wind turbines while continuing to consider green fuel production from Scenario 2. Greater energy production would create more opportunities for supplying both countries and bigger ammonia production.

When it comes to the EnergyPro model, the wind power has been increased to 4,8 GW in total. The electrolyser has a capacity of 1 061,3 MW and the ammonia synthesis unit has capacity of 525,1 MW. The O&M costs were based on data that had already been used in Scenarios 0 and 2.

All of the assumptions made in previous scenarios remain unchanged. Also there is no limit set on the amount of ammonia that can be sold.

7 Analysis of the results

7.1 Comparison

A comparison involving all of the scenarios was made, which was divided into three parts: an energy and economic comparison and an analysis from a time-based perspective. The data and assumptions presented in the previous chapter were used in EnergyPro, and the results of the simulation are displayed in a graph form.

7.1.1 Annual energy conversion summary

Firstly, an analysis is conducted of the energy utilised on the island for the purpose of ammonia production, and the energy transmitted out of it to Zealand and Germany. As illustrated in Figure 15, there are four distinct groups: the electrolyser, the ammonia synthesis unit, and exports from OEH. The initial two processes require energy to produce hydrogen, and consequently ammonia.

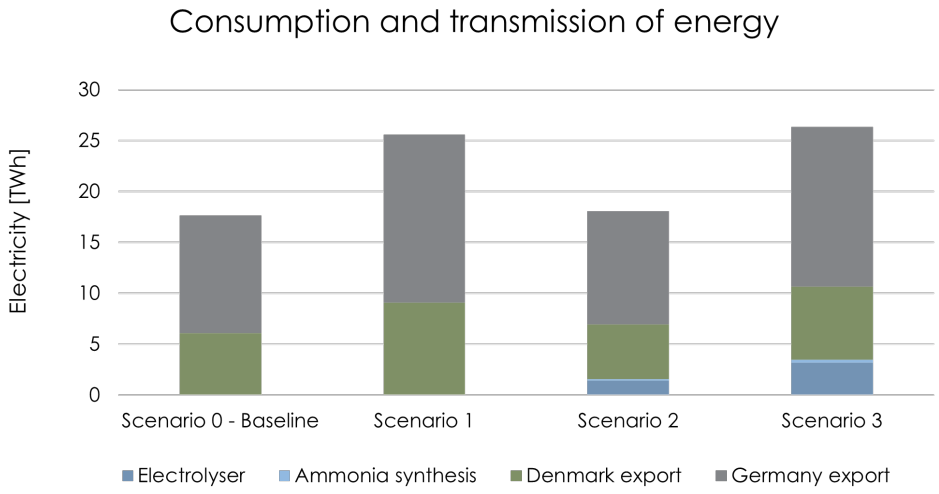


Figure 15: Consumption and transmission of energy - 2024 perspective (source: own work)

Between scenarios 0/2 and 1/3, the wind farm produced 48 % more energy, enabling more energy to be transmitted off the island. The simulation programme makes decisions based solely on the net economic costs of energy production, import to Denmark and Germany, and the operation of units producing hydrogen and ammonia for scenarios 2 and 3. In the context of the re-

spective scenarios, a comparable percentage distribution could be observed among the options, for instance, 65 % of the produced energy being allocated to Germany in scenarios 0 and 1, despite the varying energy production levels exhibited by both scenarios. For the last two scenarios, the energy used in the electrolyser and the ammonia synthesis unit is responsible for 7,81-12,09 % and 0,75-1,15 % of the energy consumption, respectively.

The subsequent discussion will focus on the exports and imports enabled by HVDC cables. The main objective is to analyse the differences between them in the context of each individual scenario, with particular reference to the level of energy production and/or PtX utilisation (see Figure 16).

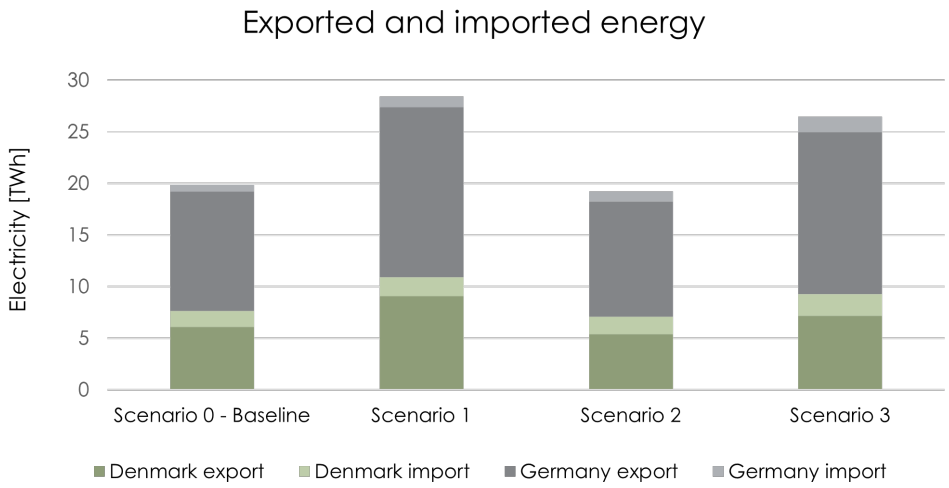
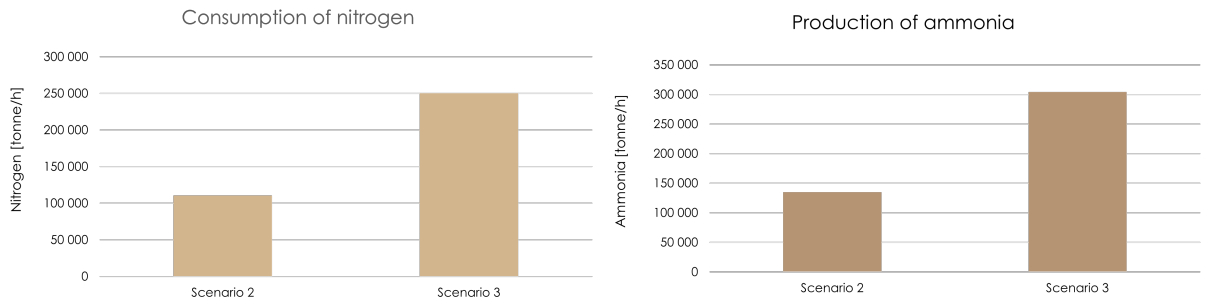


Figure 16: Consumption and transmission of energy - 2024 perspective (source: own work)

In scenarios 0 and 2, there is an export and import ratio change, respectively, and accordingly in scenarios 1 and 3. This is due to the addition of a PtX to the basic scenarios with just the energy production from the wind farm. As previously stated, the energy consumption of PtX plants is relatively low, thereby causing the changes in 4,71-8,30 % for Denmark and 3,85-4,94 % for Germany to be undetectable on the graph. From scenario 2 to 3, an increase in nitrogen demand and thus ammonia production can be observed as more energy is transferred to the electrolyser and ammonia synthesis unit. As demonstrated in Figure 17, the utilisation of 139 254 tonnes more of nitrogen has been shown to enlarge ammonia production by 26 %, resulting in a total increase of 169 409 tonnes. It does not correspond exactly to the 48 % increase in wind pro-

duction, as ammonia production is closely connected to the capacity of the units and the electricity prices distribution.



(a) Nitrogen (b) Ammonia

Figure 17: Consumption of nitrogen and production of ammonia - 2024 perspective (source: own work)

The concluding section summarises the results in terms of energy flows, as illustrated in Figure 18. Both the consumption of electricity and exports are presented with negative values, as they are regarded as outputs. The generation of energy from wind farms, in addition to the imports of energy, is regarded as a positive flow (input). As demonstrated graphically, the aggregated balance for each scenario amounts to zero, as the energy produced is wholly utilised for each of them.

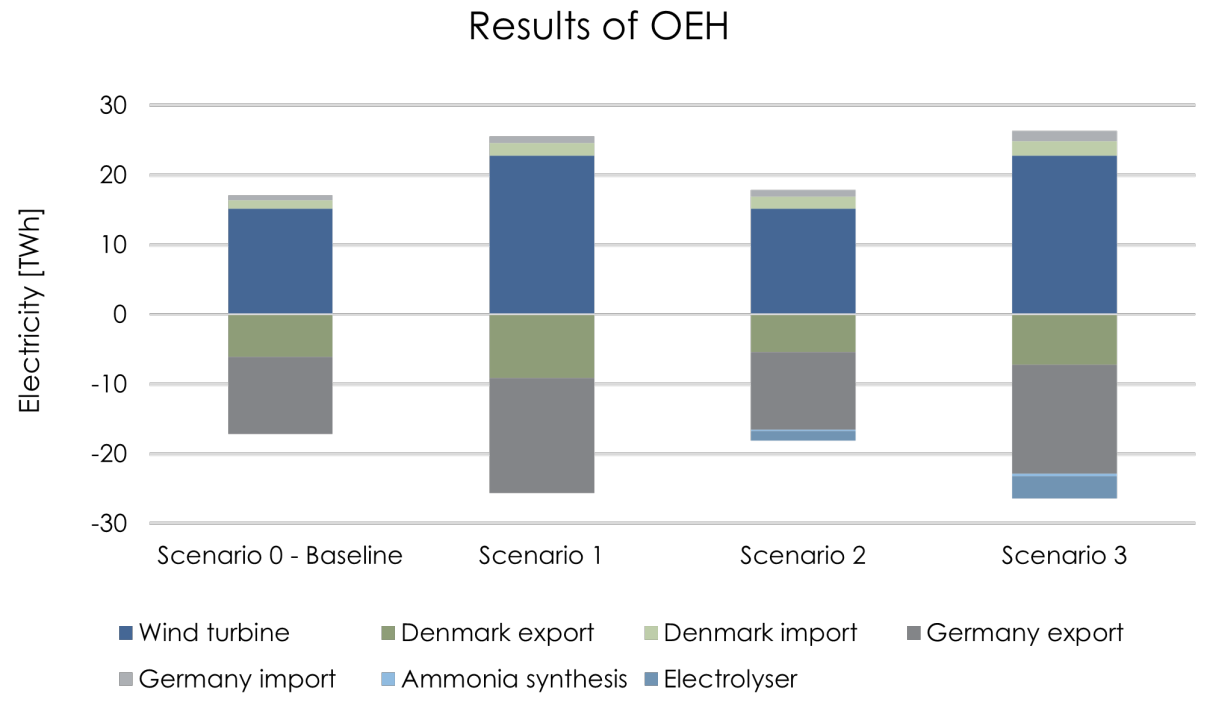


Figure 18: Results of simulation - 2024 perspective (source: own work)

7.1.2 Annual economic summary

Prior to conducting an economic analysis, it was necessary to specify certain investment costs. In all of the outlined scenarios, the values employed are presented below:

- The financial outlay required for the construction of each wind turbine was approximately €2,26 million per MW [43], inclusive of the costs associated with the wind turbine itself, its foundation, installation, wiring and other financial expenses such as sea rights fee and insurance. Furthermore, the CO_2 emissions associated with the construction would amount to 528 tonnes of CO_2 per MW [44].
- The green ammonia facility was constructed with an investment of €1,55 million per MW [48]. Due to the fact that data was provided only for 2020 and 2030, it was necessary to make an approximation.
- The investment required for the PEMEC electrolyser was estimated at €1 425 per kW, based on data for the year 2025 [48].
- Based on the already awarded contract to NKT by the Danish TSO Energinet for the Zealand connection of the OEH, the total investment of the HVDC cables could be estimated. The €650 million contract is in response to 217 km of cables (both onshore and offshore) and up to 3,8 GW of power [52]. A total of €2,5 million per GW per km can thus be assumed, with a total cable length of 390 km.

In light of the data presented above and the preceding assumptions concerning the variable and emission costs, a tabular summary could be constructed:

Table 1: Variable costs (source: own work)

Variable costs [M€/year]	Scenario 0	Scenario 1	Scenario 2	Scenario 3
O&M wind turbine	246	368	246	368
O&M ammonia synthesis	-	-	4	24
O&M electrolyser	-	-	9	15
Nitrogen	-	-	16	35
Import DK2	85	115	87	123
Import DE	21	40	22	44

Table 2: External costs (source: own work)

External costs [M€/year]	Scenario 0	Scenario 1	Scenario 2	Scenario 3
Wind farm CO_2 emissions	8	12	8	12

Table 3: Capital costs (source: own work)

Capital costs [M€]	Scenario 0	Scenario 1	Scenario 2	Scenario 3
Wind farm	7 348	11 022	7 348	11 022
Electrolyser	-	-	922	1 512
Ammonia synthesis	-	-	134	814
HVDC cable	3 120	4 680	3 120	4 680

To calculate operational income, yearly revenue must also be taken into account. The results from EnergyPro were used, including revenue from the sale of ammonia and exports to Denmark and Germany and are presented in Table 4. The programme used the price of ammonia and the hourly electricity price for both countries.

Table 4: Annual revenue (source: own work)

Annual revenue [M€]	Scenario 0	Scenario 1	Scenario 2	Scenario 3
Ammonia sale	-	-	94	213
Export DK2	419	572	400	518
Export DE	888	1 221	880	1 206

An annual financial review was then summarised, incorporating both annual revenue and outflows (variable and external costs) for the reference year 2024.

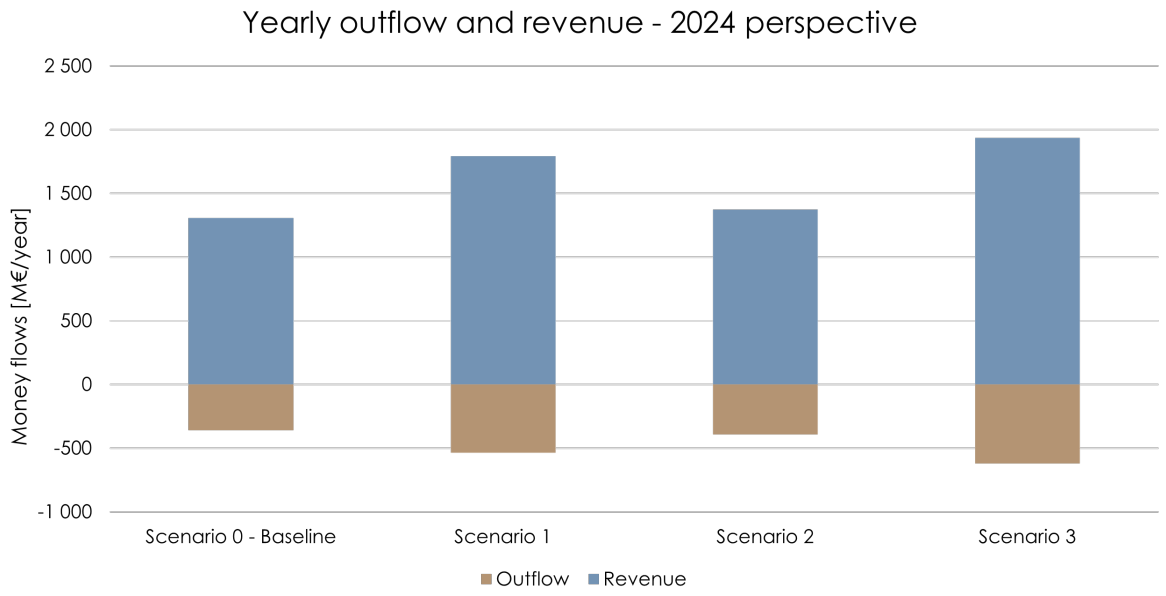


Figure 19: Yearly outflow and revenue - 2024 perspective (source: own work)

As shown in Figure 19, the annual revenue exceeds the outflows for all scenarios. Scenario 3 demonstrates the highest outflow, with €620 million, and the highest revenue, with €1 938 million. Accordingly, the most favourable performance scenario is scenario 3, with a yearly balance of €1 317 million.

Table 5: Yearly balance - 2024 perspective (source: own work)

[M€/year]	Scenario 0	Scenario 1	Scenario 2	Scenario 3
Yearly balance	947	1 259	982	1 317

7.1.3 TIME PERSPECTIVE

As is typical of investments in RES, the initial investment costs are very high. Looking at the time horizon from 2024 to 2054 with a 5 % discount rate, the 2024 present value (PV) of capital costs account for 65,5 % of total costs, while the 2024 PV of emission costs for 2024-2054 account for just 0,7 % for all scenarios. The summary is presented in the graph below:

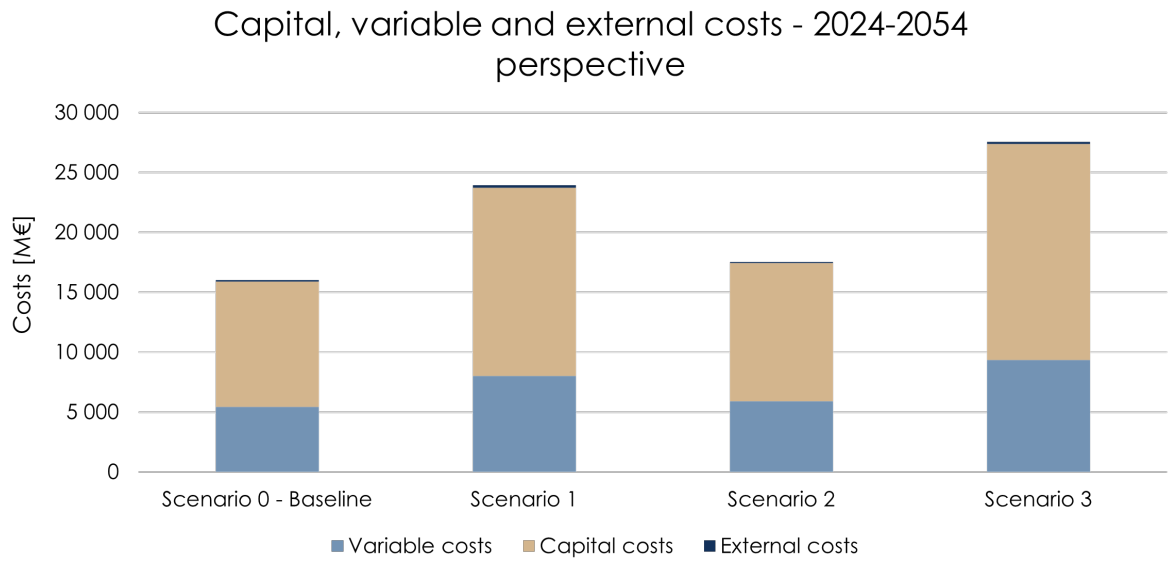


Figure 20: Capital, variable and external costs - 2024-2054 perspective (source: own work)

In order to further compare the scenarios against each other, the measure of levelised cost of electricity (LCOE) can be utilised. In the interest of the project, it was determined that the optimal approach would be to incorporate wind farm electricity production. The following formula was thus employed:

$$LCOE = \sum \frac{\frac{Inv+O\&M(t)}{(1+r)^t}}{\frac{Prod(t)}{(1+r)^t}}$$

for:

- Inv - the initial cost of investment
- O&M(t) - the yearly maintenance and operations expenditures, here O&M costs of wind farm and PtX units together with costs related to CO_2 emissions

- Prod(t) - the yearly production of energy, here electricity produced by wind farm
- r - discount rate, here 5 %
- t - the lifetime of the system, here 30 years

Since the present analysis focuses exclusively on costs rather than income and considers the time perspective, the results show a different picture to that of a simple yearly balance calculation. Discount rate of 5 % was chosen. The rate of 4 % is the one recommended by the Danish Finance Ministry for socio-economic analysis [53]. However, as the project is a long-term one and is expected to be completed not earlier than 2035, a higher rate was adopted. As illustrated in Table 6, scenario 0 exhibits the lowest LCOE, while scenario 3 reveals the highest cost, given that financial considerations incorporate the payments associated with PtX units.

Table 6: LCOE - 2024-2054 time horizon with 5 % discount rate (source: own work)

[M€]	Scenario 0	Scenario 1	Scenario 2	Scenario 3
O&M costs	254	380	232	454
Investment costs	10 468	15 702	11 524	18 028
LCOE [M€/MWh]	**58,57**	**59,41**	**64,68**	**69,00**

A net present value (NPV) analysis could be performed using investment costs and yearly balance of costs. NPV is a particularly effective metric for passive, initial investment decisions. This is because it evaluates cash flows over a longer time horizon and supports more informed investment choices [54]. The analysis employed once again a 5 % discount rate and a 30-year lifetime for the energy system. The calculation's summary is presented in table 7. The calculation enabled the determination of a payback period, which is 16 years for scenario 0, 17 years for scenario 2, 19 years for scenario 1 and 20 years for scenario 3.

Table 7: NPV - 2024-2054 time horizon with 5 % discount rate (source: own work)

[M€]	Scenario 0	Scenario 1	Scenario 2	Scenario 3
Net Present Value	4 813	4 612	4 320	3 233

7.2 Sensitivity analysis

The programme's features enabled a more in-depth analysis to be performed, examining how different inputs affect the project's most important outputs. Specific versions of the analysis were selected based on prior knowledge, with the aim of exploring unusual simulation results.

7.2.1 Electricity prices - disparity

The EnergyPRO simulation highlighted the project's significant dependence on electricity prices, in particular the difference in electricity prices between Germany and Eastern Denmark. The operating expenses stay the same, but the selling price of electricity differs following the rules of the wholesale market. The impact of the difference in prices between Germany and Denmark is explained by the fact that Bornholm is connected to both countries by HVDC cables, meaning that it can easily make profits by buying cheaply and selling more expensively when prices differ from one country to the other. This allows for a large cash inflow, since the only losses occur in terms of efficiency through transmission cable losses.

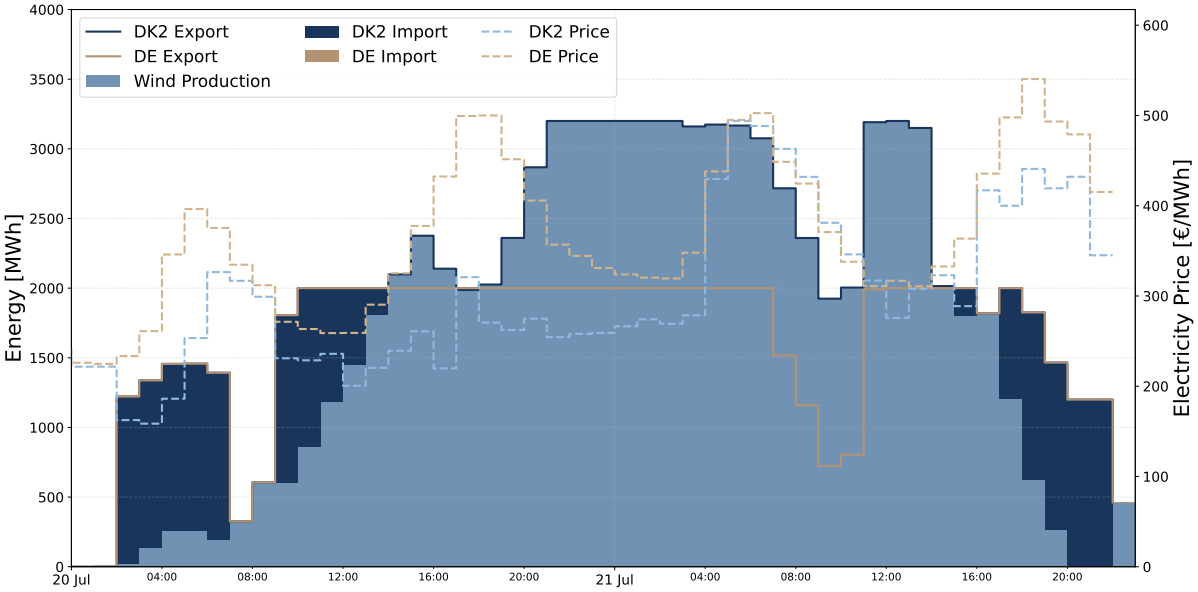


Figure 21: Evaluation of Offshore Energy Hub power exchanges and price signals between DK2 and Germany for low and high wind production conditions - time period example. (source: own work)

Figure 21 illustrates those kind of situations. The first two hours of 20th of July show a situation, where there is no electricity production from the wind farm and the prices in both countries are very similar. Electricity exchange from one country to the other at this point would result in economic losses due to transmission cable losses. Then the electricity price in DK2 decreases while the price in DE increases, creating a financial opportunity. Consequently, electricity is imported from DK2 and exported to DE. Note that import is limited by the capacity of the transmission cable from DK2 to Bornholm. Around noon on the same day, production increases, leading to imports from DK2 up to the capacity limit between DE and Bornholm. The 21st of July highlights the distribution of electricity between the two countries when electricity is being produced on Bornholm. Initially, the price in Germany is higher, leading to a priority for exporting there until the capacity is reached. Then, around 08:00, the price in DK2 becomes more advantageous, leading to a switch in priority. There is no import as soon as the link to the country with the highest electricity price reaches capacity, or when the price difference between the two countries becomes too small. The same results are observed in both 3,2 and 4,8 GW scenarios, scenario 1 will thus not be represented in this section.

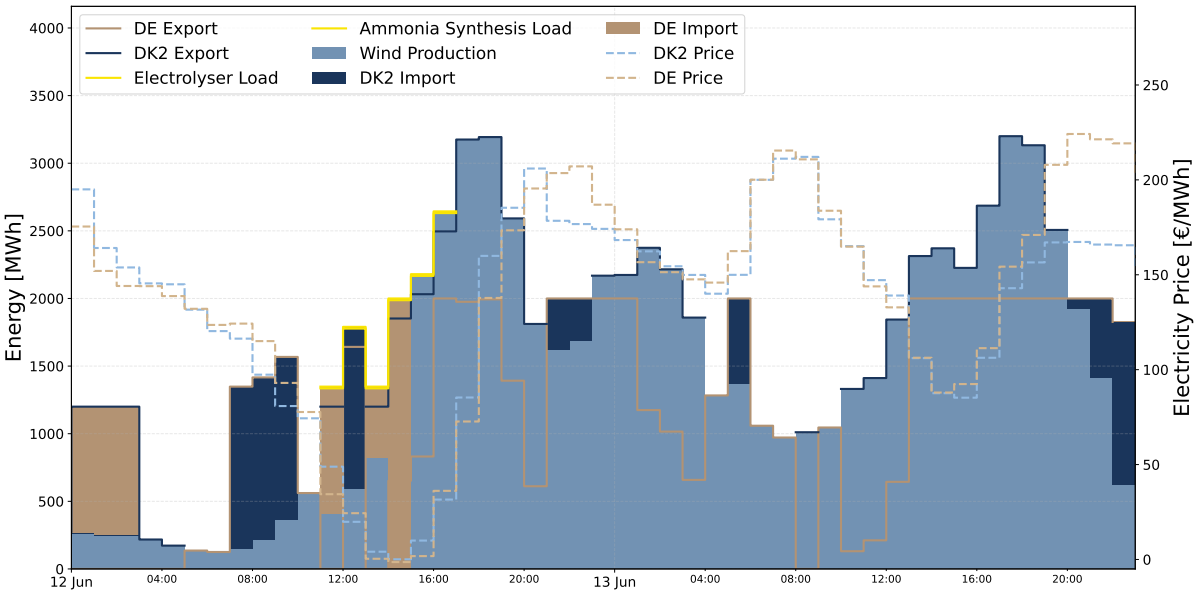


Figure 22: Evaluation of Offshore Energy Hub power exchanges and price signals between DK2 and Germany for low and high wind production conditions and ammonia production - time period example. (source: own work)

As illustrated in Figure 23 the behaviour is comparable to the one of the scenarios without PtX for the majority of the time. However, the OEH will prioritise the production of ammonia when the price of electricity in one of the countries goes below the production cost of ammonia. This decision will be made by the programme, based on a comprehensive analysis of nitrogen, ammonia and electricity prices.

Lower electricity prices exert a direct influence on ammonia production by enabling the PtX units to operate competitively. In practice, this means that PtX assets function more as a stabilising source of revenue for the OEH rather than its primary income stream.

7.2.2 ELECTRICITY PRICES - DEVELOPMENT

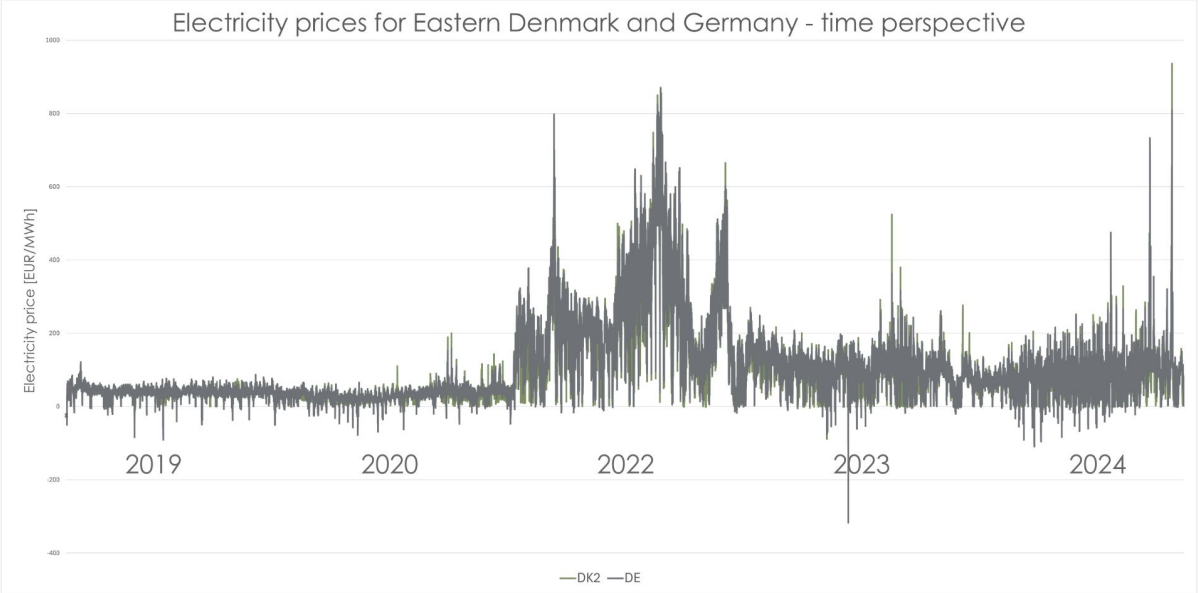


Figure 23: Change of electricity prices throughout the years for Eastern Denmark and Germany (source: own work)

As mentioned previously in section 7.2.1, lower electricity prices greatly affect the work of PtX. When electricity prices approximate those witnessed in 2019 or 2020, ammonia production levels are high and consistent. Conversely, under price conditions analogous to those experienced in 2022, there is a sharp decline in production, although it persists while the main revenue is from selling the electricity to Germany. In addition, it should be noted that in years with low energy prices, the NPV remains in the negative after 30 years. Furthermore, ammonia sales are particularly significant, especially in scenario 3, where they exceed both the sales of electricity to Germany and to Zealand.



Figure 24: Consumption and transmission of produced energy - scenario 3 in different years (source: own work)

7.2.3 Cost of CO_2 emissions

As the price of CO_2 is not incorporated within the programme's simulation, it will not impact the operation of the OEH. However, it will influence the overall annual economic income. The calculation was conducted for a range of prices, beginning at €55 and extending to €95 per tonne of emitted CO_2. The average proportion of external cost to operation income varies between 0,65 % and 1,12 %, which is not a significant amount compared to variable costs. Furthermore, small changes in LCOE are evident, with value changes remaining within a range of €0,56 per MWh. Payback time alterations are not evident for any of the options.

Table 8 presents a detailed breakdown of the specific changes in the net present value between the various scenarios and the CO_2 prices. As shown in the table, the value is at its highest for scenario 0 and lowest for scenario 4. The overall percentage value change is less than 3,2 % from the baseline value of €75.

Table 8: Change of NPV - 2024-2054 time horizon with 5 % discount rate - based on CO_2 prices (source: own work)

[M€]	Scenario 0	Scenario 1	Scenario 2	Scenario 3
55 €/t Co_2	4 882	4 714	4 403	3 214
65 €/t Co_2	4 847	4 663	4 369	3 163
75 €/t Co_2	**4 813**	**4 612**	**4 335**	**3 233**
85 €/t Co_2	4 779	4 560	4 301	3 061
95 €/t Co_2	4 744	4 509	4 266	3 010

7.2.4 PtX

Capacity

To evaluate how the installed PtX capacity influences the project economics, several capacity levels were assessed. The baseline configuration corresponds to the capacity derived from the EnergyPRO optimisation, which identifies the ammonia plant size that maximises revenue based on the hourly profiles of electricity consumption, ammonia synthesis, electrolysis demand, and electricity prices in both countries. In practice, the optimisation determines the configuration that yields the highest value per hour, either by selling electricity to the grid or by converting it into ammonia, trough one whole operation year.

This optimisation results in an installed ammonia synthesis capacity of 86,25 MW_{NH_3}, corresponding to an hourly maximum output of approximately 16,6 tonnes of ammonia. Most of the electricity demand originates from water electrolysis, which produces the hydrogen required for subsequent ammonia synthesis, while the synthesis itself consumes comparatively little electricity.

The electrolyser capacity was then set to the minimum value that allows full utilisation of the optimised ammonia synthesis capacity. This was obtained by initially oversizing the electrolyser and progressively reducing its capacity until reaching the threshold where the electrolysis would become the limiting production unit. In the base scenario, this results in an electrolyser capacity of 174,7 MW_{el}. These capacities correspond to utilisation factors of 27 % for ammonia synthesis and 32 % for electrolysis.

A sensitivity analysis was conducted on the impact of different PtX capacity levels, with results presented in Table 9. The outcomes show that the Net Present Values remain within a narrow range: higher ammonia output is systematically offset by the increased investment cost of larger installations. Overall, the economic effect of scaling the PtX units is limited, but possible.

As discussed in Section 7.2.1, ammonia production increases significantly under low electricity price conditions. This implies that in scenarios where electricity becomes cheaper in the future, consistent with EU objectives for a com-

petitive and affordable power market, ammonia production could become a more attractive revenue source. At the same time, PtX retains a high operational flexibility, securing income even during periods of very low or negative electricity prices.

Table 9: Net Present Value for different PtX capacities (2024–2054, 5% discount rate) (source: own work)

PtX capacity scenario	0	Low	Medium	High	Very high
Electrolyser capacity [MW]	0	87,4	174,7	349,4	873,6
Ammonia synthesis capacity [MW]	0	43,1	86,3	172,5	431,3
NPV [M€]	4 813	4 759	4 707	4 602	4 282

Price of nitrogen and ammonia

As seen in table 10, the NPV of the project increases more than linearly with rising ammonia prices. As ammonia becomes more valuable, the OEH operation shifts: electricity imports increase whenever electricity prices are lower than the marginal revenue gained from producing ammonia, and electricity exports decrease as the OEH more frequently prioritises ammonia production.

Table 10: Net Present Value for different ammonia prices (2024–2054, 5 % discount rate) (source: own work)

Ammonia price [€/kg]	0.35	0.70	0.85	1.05
Scenario 0	4 813	4 813	4 813	4 813
Scenario 2	4 552	4 707	4 820	5 027

Despite this shift, the share of electricity exported remains sufficiently high, meaning that the OEH continues to fulfil its primary role as a safe, affordable and competitive electricity producer.

It is observed that the same NPV as in scenario 0 is reached at an ammonia price of approximately 0,85 €/kg. This value is indicative, since it is calculated using the electricity price distribution of the year 2024, and the resulting NPV depends on the average electricity prices of the considered period. Nevertheless, higher ammonia prices, or higher revenues from ammonia sales, reduce the uncertainties associated with electricity price volatility.

7.2.5 Losses in the HVDC cables

Losses in the HVDC cables connecting the OEH to both countries limit the effective transfer of energy. As losses increase, transporting electricity becomes less attractive whenever the price difference between the two countries narrows. This primarily reduces imports, since exports from the OEH remain viable as long as electricity prices stay above a certain threshold. It is also observed that, under higher loss conditions, ammonia production becomes more favourable, since it is not affected by transmission losses when wind generation is available.

As expected, higher losses result in lower transmitted volumes and a corresponding reduction in project NPV. This emphasises the importance for any energy project of minimising all forms of system losses.

In scenario 2, the decrease is approximately linear and corresponds to an NPV reduction of about 30 M€ per year for each additional percentage of cable losses.

8 Discussion

Following the formulation of the primary project objectives, a shift in perspective was observed. The project, initially motivated by the objective of achieving climate neutrality and expanding RES, has evolved into a prospective pillar of Europe's security and resilience. Consequently, it has become a strategic necessity for the continent's future, as outlined in this report, which is anchored in both economic and technical viability analysis.

8.1 Technical challenges

As shown in the previous chapter, the analysed scenarios are technically feasible within the assumed conditions and modelling framework. However, the simulations also revealed several technical challenges that need to be addressed to ensure that the OEH operates reliably and efficiently.

While a PtX integration is technically feasible, it introduces operational complexity to the BEI. Although the PEMEC electrolysis and ammonia synthesis process can operate reliably with variable wind input, low utilisation factors (27–32%) and conversion losses limit its role in the initial phase (Section 7.2.4). The electrolyser efficiency varies by approximately ±5% depending on input power variability, degrading performance under the highly fluctuating conditions typical of offshore wind (Section 6.1.4). This technical constraint means that PtX functions as a flexible, storage-enabled extension that stabilises system operation during periods of low electricity prices rather than acting as a primary revenue driver. Additionally, HVDC transmission losses of 3 % reduce available energy, and expanded wind capacity scenarios require larger or additional cable systems, further increasing technical complexity in infrastructure design and cross-border grid coordination.

Despite current technical and operational constraints, PtX is key to the long-term decarbonisation of Europe's energy system. Facilities at OEHs can convert surplus renewable electricity into green hydrogen and ammonia, which can then be used to supply hard-to-abate sectors such as shipping, and aviation,

which cannot be directly electrified. As the Energy Island Roadmap emphasises, OEHs integrated with PtX capacity will become a cornerstone of future energy systems by 2050. These hubs can provide decarbonised energy carriers. These carriers will support sector coupling and will also enable a fully renewable European energy infrastructure. Bornholm's configuration offers a valuable demonstration site for testing PtX technologies and operational strategies that will be crucial for achieving the EU's climate neutrality targets beyond 2030. This is particularly relevant given that technological advances and cost reductions will improve the economic viability of these technologies over the coming decades [55].

8.2 Cost-effectiveness

In accordance with the economic results that have been presented in the preceding subchapters (Sections 7.1.2 and 7.1.3), the project has been demonstrated to generate favourable monetary outcomes for all the scenarios. Therefore, the project should be deemed to have the green light for progression.

The findings of this analysis indicate that the implementation of wind turbines alone would result in more financially viable scenarios. It is evident based on the LCOE that with each expansion, the cost of energy production will increase (see Table 6). This conclusion is also supported both by the NPV values after a 30-year period and by a payback time that is three years less in comparison with the respective scenarios involving the expansion with additional PtX (see Table 7). However, the observation that scenario 0 and scenario 2, with the addition of PtX to the 3,2 GW wind farm, exhibits similar NPV values, suggests that the investment cost of the wind turbines itself is responsible for a significant proportion of the overall final results. Furthermore, a discrepancy of approximately 300 M€ can be observed in the yearly revenue between the given scenarios (see Table 5). This underscores the substantial significance of investment costs, which are of such magnitude that the transition towards PtX appears unfavourable.

Accordingly, it could be concluded that if investment costs were to decrease, scenarios involving PtX technologies could prove to be more economically vi-

able. This assertion may also be applied to the concept of wind farm expansion, as such a development could become financially viable if it were to result in additional income from PtX. It is possible that the investment cost may be reduced, as the calculations were conducted on the basis of prices quoted or estimated for 2024/2025, and the construction of equipment related to the production of hydrogen or ammonia is not planned in the initial phase of the project. It is important to note this project represents a first of its kind in Europe, which led to the adoption of rigorous approach, including the implementation of a higher discount rate than recommended. The technology is not currently mature, as evidenced by the fact that, for example, taking the investment data for PtX for 2040, it can be seen that scenarios 0 and 2 could achieve identical NPVs.

In summary, the OEH on Bornholm is an investment project that holds considerable appeal. Although there is a significant amount of energy used in the electrolyser and the ammonia synthesis unit, a greater quantity of energy is being exported, as it is more advantageous for the OEH to sell the energy to other countries. This highlights the potential economic role of PtX in supporting the island in case electricity prices in both countries are not attractive. The additional prospective benefits of oxygen sales, potential for a reduction in investment costs through optimisation of the PtX capacity and confirmed financial support from the EU, which are not included in the analysis, confirm that the PtX extension could have even more significant potential.

8.3 Risks and uncertainties

This section discusses the main sources of uncertainty affecting the OEH, with a particular focus on economic and market-related risks.

8.3.1 Sensitivity analysis

The sensitivity analysis shows that the economic performance of the OEH is mainly exposed to market uncertainties, rather than to purely technical parameters. While all tested variables influence the NPV, their impact differs in both magnitude and nature.

Electricity prices (Sections 7.2.1 and 7.2.2)

Electricity prices are the most influential source of uncertainty. Differences between prices from Zealand and Germany directly affect the profitability of cross-border electricity exchanges. When price spreads are high, the OEH benefits from arbitrage opportunities and electricity exports become a key revenue stream. Conversely, when price convergence occurs, electricity is no longer imported for export to another country. Instead, electricity production is either exported or used for internal conversion, such as PtX, based on how low electricity prices are compared to the benefits of producing ammonia.

In addition to spatial price differences, long-term price development introduces further uncertainty. Higher average electricity prices increase export revenues but are often associated with higher volatility. This volatility increases the value of flexibility within the system, as PtX can absorb surplus electricity during low-price periods. As a result, electricity price uncertainty affects not only revenues, but also the preferred operational strategy of the OEH.

Ammonia price (Section 7.2.4)

The ammonia price represents another major economic uncertainty. Higher ammonia prices progressively shift system operation away from electricity exports toward ammonia production. This shift does not occur abruptly, but rather gradually, as the system balances export opportunities with conversion constraints. Furthermore, ammonia sales provide a revenue stream that is less directly linked to electricity market volatility, thereby improving income stability. At sufficiently high ammonia prices, PtX becomes a central contributor to project value rather than merely a flexibility option.

PtX capacity (Section 7.2.4)

Uncertainty related to PtX capacity sizing mainly concerns capital efficiency. Increasing PtX capacity improves the ability to exploit low-price periods, but the analysis reveals diminishing returns beyond a certain threshold. Once most surplus electricity is already captured, additional capacity remains underutilised,

increasing investment risk. This highlights the importance of aligning PtX capacity with realistic price volatility and expected ammonia demand.

CO_2 price (Section 7.2.3)

The CO_2 price has a more limited direct impact on the NPV. While it affects operating costs, its influence is largely transmitted indirectly through electricity prices, as higher CO_2 prices raise fossil-based marginal generation costs. Although its short-term economic effect is moderate, CO_2 pricing strengthens the long-term competitiveness of renewable-based solutions and supports the robustness of the OEH business case under stricter climate policies.

HVDC losses (Section 7.2.5)

HVDC transmission losses introduce a technical uncertainty with relatively small economic consequences. Higher losses reduce net export revenues, but their overall impact remains modest compared to market-driven factors. Losses become more relevant in scenarios dominated by electricity exports, where they slightly reduce profitability and indirectly favour on-site conversion through PtX.

8.3.2 Stakeholders and community acceptance

These technical and economic constraints do not exist in a vacuum. The actors involved in the BEI project, from European institutions to local communities, each have distinct interests and influence over which scenarios materialise. Understanding this stakeholder landscape clarifies why certain pathways may be favoured or delayed despite their apparent economic viability.

According to standard definitions, a stakeholder is any group, person, or organisation that has an interest in the project's development, outcome, and impact [56]. Identifying different stakeholders makes it possible to see the bigger picture and assess which parties benefit most, which ones hold decision-making power, and whose concerns should be prioritised. In a large infrastructure project like BEI, numerous stakeholders with different perspectives and interests are

involved; some directly, and others indirectly. Considering all of them during the planning phase is essential for maximising overall project success while mitigating potential negative implications.

The stakeholder matrix (see Figure 25) shows 14 different stakeholders in total. The matrix is structured along two dimensions: influence (the ability to affect the project's outcomes) and interest (degree to which they are affected by or concerned with the project). Stakeholders were first identified from project documents, information gained at the conference, and knowledge of the Bornholm energy context, and then qualitatively assessed and positioned according to their expected influence and interest. The assessments were made in a group discussion, drawing on examples from the stakeholder theory covered during classes and our understanding of each stakeholder's formal role and responsibilities in the energy system. This results in stakeholders positioned on the matrix in either of those four categories: Manage Closely, Keep Satisfied, Keep Informed, or Monitor. Based on their location on the matrix and general profile, they can be grouped for easier comparison.

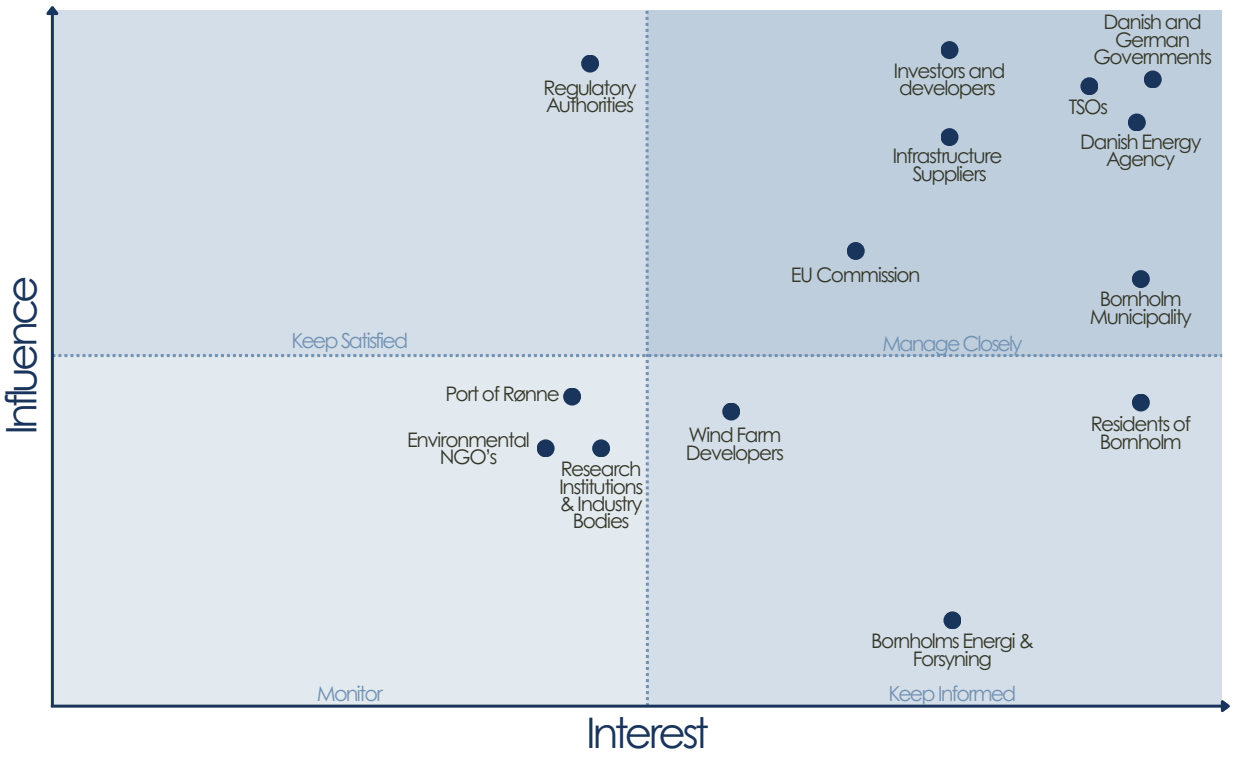


Figure 25: Stakeholders matrix (source: own work)

European and international institutions

The EU Commission appears as a relatively influential but slightly less directly involved stakeholder, whose main interest lies in cross-border integration, security of supply, and meeting EU climate targets. By granting funding and emphasising the project's importance, the European Commission shapes this investment's financial viability and pushes for the hybrid interconnector design that links Denmark and Germany through Bornholm. German and Danish governments are mapped at the top-right of the grid, as their formal agreements on capacity, grid connections, and environmental rules are the foundations for the project to move ahead.

Public authorities

The Danish Energy Agency and TSOs Energinet and 50Hertz are all at the high-influence, high-interest corner because they design the tenders, grid concept and regulatory framework for the energy island and its offshore wind farms. Their main concern is to meet climate, security, and supply goals while ensuring a cost-effective, technically robust solution, so they must be managed very closely in any project strategy. Every change or idea should be discussed and examined with them. Bornholm Municipality also has relatively strong influence, as it coordinates local planning approvals and impacts island's community (and therefore, local acceptance), linking national ambitions with local development needs. Because of the project's national and international importance, the influence of this specific municipal governance is lesser than that of higher-level governments.

Market and infrastructure actors

Wind farm developers such as Ørsted, Vestas, Siemens Gamesa and Vattenfall have a strong commercial interest in the construction and operation of turbines, and related offshore infrastructure. Their involvement is motivated by market opportunities and the potential to secure installation or long-term O&M contracts. However, their formal decision-making power in shaping the project

is relatively limited, as turbines and services can typically be sourced from a competitive global market. While these companies are important partners, the project is therefore not structurally constrained by any single turbine developer.

By contrast, infrastructure suppliers such as Siemens Energy or Hitachi occupy a more influential position, as they provide specialised high-voltage equipment, converter technology and other critical components that must be tailored to the hybrid interconnector and energy-island concept. Their technological capabilities and delivery timelines can significantly affect project feasibility, sequencing and risk, placing them higher on both the interest and influence scales.

Investors and developers, such as Copenhagen Infrastructure Partners and RWE, show a similar level of influence and interest. Their focus lies in long-term, stable returns, and they require certainty regarding revenue frameworks, regulatory stability, and risk allocation before committing capital. Their financing decisions can accelerate or delay project milestones, giving them an important gatekeeping role.

Lastly, the local Port of Rønne functions as a logistics hub that will benefit from increased economic activity and supply-chain participation, with future potential as a fuelling port for ammonia-powered ships. Bornholms Energi & Forsyning, the local utility company, has high interest but relatively low influence, as it manages only the local grid, which lies outside the core energy-island infrastructure. It should therefore be kept informed, but is not a central decision-maker in the project.

Local community and civil society

Residents of Bornholm are highly interested in the project because it affects their jobs, landscape, environment, and quality of life, but individually they have less formal power and their influence could be expressed through local elections, participation in consultations, and other forms of activism. Maintaining trust, sharing benefits, and addressing concerns about noise, visual impact

or marine use conflicts is essential, which is why they sit in the "manage closely" quadrant despite relatively modest institutional power.

Environmental Non-Governmental Organisations (NGOs) and the Danish Society for Nature Conservation focus on protecting marine ecosystems and promoting sustainable design; therefore they are placed in the "monitor/keep satisfied" zone but can still exert meaningful influence via advocacy, consultation responses and legal challenges if environmental standards are not respected.

Interestingly, implementing this project could have a positive impact on the local environment. Artificial structures such as turbine foundations or cable protections often act as artificial reefs, providing additional hiding places, feeding grounds and spawning habitat for fish and invertebrates, and in some cases may increase local biodiversity and biomass in the immediate vicinity of the structures [57, 58].

Knowledge and innovation actors

Research institutions such as DTU, University of Southern Denmark (SDU) and AAU show medium interest and moderate influence, contributing expertise on system design, environmental impacts, PtX options, and workforce development. Their role is to reduce technological and planning risks, provide independent analysis, and support innovation that makes the energy island a replicable model for other regions. Engaging these stakeholders ensures that decisions are evidence-based and that local businesses and workers can acquire the skills needed to participate in the emerging offshore wind and green fuels value chains.

Industry bodies, such as Green Power Denmark, along with Danish Energy Industries, act mainly as advisory and advocacy bodies, providing industry expertise, shaping policy discussions and supporting supply-chain and workforce development. Like research institutions, they have moderate influence and interest: they inform decisions and reduce risks but do not hold formal authority over approvals, financing or project design.

Connection with techno-economic results

The techno-economic results presented in Chapter 7 help explain why different stakeholders may support or oppose certain development pathways. For public authorities such as Energinet, the Danish Energy Agency and national governments, scenarios with lower costs would probably be preferable, since they are easier to justify within regulatory and political frameworks. The results also show that PtX mainly acts as a flexibility option rather than a main source of revenue under current assumptions. Its economic performance strongly depends on ammonia prices and electricity market conditions. This would make private investors and developers have a more cautious approach, because they need stable returns and clear risk allocation before committing capital. In contrast, strategic actors such as the European Commission and national governments may still support larger-scale or PtX-based scenarios, as these options contribute to long-term goals such as cross-border energy integration, future infrastructure development and the creation of green fuel markets, even if short-term economic results are weaker.

Overall, the stakeholder analysis shows that the future development of the BEI project depends not only on technical and economic performance, but also on how different actors understand risks, benefits, and long-term value. The economic results help clarify why some scenarios are more attractive to regulators and investors, while others rely more on political support and strategic ambitions. Aligning stakeholder interests therefore requires balancing economic efficiency with broader goals such as energy security, innovation, and climate leadership, supported by transparent decision-making, and early stakeholder engagement.

9 Conclusion

This analysis evaluates the feasibility of the BEI project by examining four development scenarios, which range from the approved baseline of 3,2 GW of wind capacity to an expansion involving an additional 1,6 GW of wind capacity and PtX integration. It directly addresses the effect of different development pathways on the project's technological configuration and economic viability.

The technical analysis shows that all four scenarios are technically feasible. However, PtX introduces operational complexity, technical constraints, and substantial additional infrastructure leading to low utilisation factors of 27-32 %. Additionally, HVDC transmission losses of 3 % reduce available energy, and expanded wind capacity scenarios require larger or additional cable systems, further increasing technical complexity in design and crossborder grid coordination.

The results of the economic analysis indicate that the wind-only scenarios would outperform the PtX scenarios. Incorporating time perspective and discount rate of 5 %, the payback period for the considered scenarios varies between 16 and 20 years, situated within the technical lifetime of the energy system of 30 years. The potential economic role of PtX in supporting the BEI, in combination with the possibility of additional incomes not incorporated into the analysis, supports the viability of PtX extension.

PtX mainly acts as a flexibility option rather than a primary revenue driver, with its economic relevance strongly dependent on ammonia prices and electricity price distributions. At sufficiently high ammonia prices (above 0,85 €/kg), PtX shifts from a marginal flexibility role toward a more significant contributor to system value. Should investment costs be reduced in line with future projections, there is a strong possibility that PtX scenarios will become competitive.

The stakeholder analysis reveals that European and national authorities, TSOs and the Danish Energy Agency wield significant influence over decision-making processes, prioritising cost-effective, secure and climate-aligned solutions. Wind developers and local utilities have significant commercial and operational in-

terests but limited structural influence. Conversely, Bornholm Municipality and residents are profoundly affected by local environmental and social impacts, necessitating their close engagement to ensure acceptance. Research institutions and industry bodies are responsible for knowledge and innovation support, wielding a moderate influence in this capacity. Once the BEI project enters the implementation phase, the needs and expectation of each of those stakeholders will have to be taken into consideration to ensure a just, beneficial, and socially and environmentally responsible outcome.

The implementation of an OEH on Bornholm, whether it concerns a wind farm or an additional PtX facility, is a highly promising initiative. Despite the necessity for the establishment of certain limitations, the report includes the most significant present prospects for the OEH and provides a solid foundation for future analysis. In future research, it will be important to consider these limitations, as well as the time frame, which may have a significant impact on the final results, as demonstrated by the analysis.

References

[1] Agency, I. E. *World Energy Outlook 2024*. Tech. rep. International Energy Agency, 2024. url: https://iea.blob.core.windows.net/assets/140a0470-5b90-4922-a0e9-838b3ac6918c/WorldEnergyOutlook2024.pdf (visited on 09/12/2025).

[2] BP. *BP Energy Outlook 2025*. Tech. rep. BP, 2025. url: https://www.bp.com/content/dam/bp/business-sites/en/global/corporate/pdfs/energy-economics/energy-outlook/bp-energy-outlook-2025.pdf (visited on 09/12/2025).

[3] *The energy world is set to change significantly by 2030, based on today's policy settings alone*. 2023. url: https://www.iea.org/news/the-energy-world-is-set-to-change-significantly-by-2030-based-on-todays-policy-settings-alone (visited on 09/12/2025).

[4] Roser, M. and Ritchie, H. *How has world population growth changed over time?* 2023. url: https://ourworldindata.org/population-growth-over-time (visited on 09/12/2025).

[5] *Energy supply*. 2026. url: https://www.iea.org/world/energy-mix (visited on 01/01/2026).

[6] *Designing Energy Efficiency Policies to Enhance Affordability*. International Energy Agency, June 2025.

[7] European Commission. *The European Green Deal*. 2019.

[8] European Commission. *Directive (EU) 2018/2001 of the European Parliament and of the Council of 11 December 2018 on the promotion of the use of energy from renewable sources*. 2018.

[9] European Commission. *REPowerEU: Joint EU action for more affordable, secure and sustainable energy*. 2022.

[10] Bundesregierung. *Climate Change Act - climate neutrality by 2045*. 2021.

[11] International Energy Agency. *Energy Strategy 2050*. 2021.

[12] *Publication of results from the preliminary site investigations – Offshore wind Baltic Sea*. 2026. url: https://ens.dk/en/energy-sources/offshore-wind-power/preliminary-site-investigations-energy-islands/publication-0 (visited on 01/01/2026).

[13] Lund, H., Østergaard, P. A., Connolly, D. and Mathiesen, B. V. 'Smart energy and smart energy systems'. In: *Energy* 137 (2017).

[14] Geidl, M. and Andersson, G. *A modeling and optimization approach for multiple energy carrier power flow*. Tech. rep. Swiss Federal Institute of Technology (ETH) Zurich, 2005.

[15] Østergaard, J., Christensen, E. D., Halsnæs, K., Riisager-Simonsen, C., Lisbjerg, D., Jensen, A. D., Duus, J. Ø., Madsen, H., Frandsen, H. L. and Savaghebi, M. *Denmark as the Energy Island Pioneer*. Technical University of Denmark (DTU), 2023.

[16] The Danish Ministry of Climate, E. and Utilities. *Climate Programme 2020 - Denmarks Mid-century, Long-term Low Greenhouse Gas Emission Development Strategy*. 2020.

[17] *The Danish Energy Agency publishes Plan for Program Bornholm Energy Island*. 2024. url: https://ens.dk/en/press/danish-energy-agency-publishes-plan-program-bornholm-energy-island#:~:text=Plan%20for%20Program%20Bornholm%20Energy%20Island%20sets%20the,well%20as%20high-voltage%20installations%20on%20Bornholm%20and%20Zealand. (visited on 09/12/2025).

[18] International Energy Agency (IEA). *Denmark 2023 Energy Policy Review*. Tech. rep. International Energy Agency, 2023. url: https://www.iea.org/reports/denmark-2023.

[19] Singlitico, A., Campion, N. J. B., Münster, M., Koivisto, M. J., Cutululis, N. A., Suo, C. J., Karlsson, K., Jørgensen, T., Waagstein, J. E. and Bendtsen, M. F. *Optimal placement of P2X facility in conjunction with Bornholm energy island*. 2020.

[20] Energinet. *Business case for Bornholm energy island's electricity infrastructure*. 2022.

[21] Grøn Energi, N. C. for. *Bornholm as a test and demonstration site for energy island technologies*. 2023.

[22] Mathiesen, J. K., Thostrup, M. and Petersen, S. C. 'Bornholm - a power-to-X island'. MA thesis. Aalborg University, 2023.

[23] Hedetoft, A., Hansen, A. S., Lindahl, J., Rasmussen, J. N. and Marschner, G. H. *Socioøkonomisk analyse. Lokaløkonomiske og beskæftigelsesmæssige effekter på Bornholm i forbindelse med energiøen*. Center for Regional- og Turismeforskning (CRT), 2023.

[24] Lüth, A. and Keles, D. 'Risks, strategies, and benefits of offshore energy hubs: A literature based survey'. In: *Renewable and sustainable energy reviews* (2024).

[25] Hansen, A. S., Manniche, J. and Larsen, K. T. 'Navigating sustainable transition processes at the local level: The case of Energy Island Bornholm'. In: *Environmental Innovation and Societal Transitions* (2024).

[26] Mörchen, J., Røpke, J., Nielsen, K. and Biørn, T. 'Bornholm Energy Island - Interconnecting Countries'. not published semester project from AAU.

[27] Manniche, J. and Hansen, A. S. *Test Island Bornholm 2.0. Perspectives for Bornholm as a test and development environment for green energy and technology*. Center for Regional- og Turismeforskning (CRT), 2025.

[28] *Bornholm*. 2025. url: `https://en.wikipedia.org/wiki/Bornholm` (visited on 15/11/2025).

[29] *NKT to supply power cable link for Bornholm Energy Island in Denmark*. 2025. url: `https://www.windtech-international.com/projects-and-contracts/nkt-to-supply-power-cable-link-for-bornholm-energy-island-in-denmark` (visited on 09/12/2025).

[30] *Energy Island Bornholm*. 2025. url: `https://www.energiobornholm.dk/en` (visited on 09/12/2025).

[31] *Potentials and perspectives for the island of Bornholm*. 2025. url: `https://www.energiobornholm.dk/en` (visited on 04/01/2026).

[32] *Energy Island Summit 2025*. 2025. url: https://balticenergyisland.com/energy-island-summit-2025/ (visited on 09/12/2025).

[33] Haeberle, F., Parolin, G. and Pigosso, D. C. 'Scenario building guidelines for sustainable innovation'. In: *Proceedings of the Design Society* 4 (2024), pp. 1289–1298.

[34] Habibic, A. *Bornholm to become green fuel hub for 60,000 ships*. 5th Apr. 2022. url: https://www.offshore-energy.biz/bornholm-to-become-green-fuel-hub-for-60000-ships/ (visited on 09/12/2025).

[35] *NREL Turbine Archive*. 2020. url: https://nrel.github.io/turbine-models/IEA_15MW_240_RWT.html (visited on 09/12/2025).

[36] *NKT awarded the power cable connection between Bornholm Energy Island and Zealand in Denmark*. 2025. url: https://www.nkt.com/news-press-releases/nkt-awarded-the-power-cable-connection-between-bornholm-energy-island-and-zealand-in-denmark (visited on 09/12/2025).

[37] Bellotti, D., Rivarolo, M. and Magistri, L. 'A comparative techno-economic and sensitivity analysis of Power-to-X processes from different energy sources'. In: *Energy Conversion and Management* 260 (2022).

[38] Sood, S., Prakash, O., Boukerdja, M., Dieulot, J.-Y., Ould-Bouamama, B., Bressel, M. and Gehin, A.-L. 'Generic Dynamical Model of PEM Electrolyser under Intermittent Sources'. In: *Energies* 13.24, 6556 (2020).

[39] Opeyemi A, O., Sharif F, Z. and Bing-Jie, N. 'Green ammonia production technologies: A review of practical progress'. In: *Journal of Environmental Management* (2023).

[40] Gokcay, B., Thi Tuyet, N. P., Ebru, S.-B. and Cagatay, I. 'A roadmap to alternative fuels for decarbonising shipping: The case of green ammonia'. In: *Research in Transportation Business Management* (2024).

[41] Ye, D. and Edman Tsang, S. C. 'Prospects and challenges of green ammonia synthesis'. In: *Nature Synthesis* (2023).

[42] *Electricity Maps*. 2026. url: https://www.electricitymaps.com/ (visited on 01/01/2026).

[43] *Calculation: Wind Farm*. EMD International A/S, Calculated in windPRO 4.2.271 on 24/11/2025.

[44] Vestas Wind Systems A/S. *Life Cycle Assessment of Electricity Production from an offshore V236-15 MW Wind Plant*. 2024.

[45] *EU Carbon Permits*. 2026. url: https://tradingeconomics.com/commodity/carbon (visited on 01/01/2026).

[46] *HVDC Submarine Power Cables in the World*. European Commission. Joint Research Centre. Institute for Energy Transport, 2015.

[47] Danish Energy Agency. *Technology Data – Renewable fuels*. 2025.

[48] Danish Energy Agency. *Technology Data for Energy Carrier Generation and Conversion*. 2025.

[49] *Higher Calorific Values of Common Fuels: Reference Data*. 2003. url: https://www.engineeringtoolbox.com/fuels-higher-calorific-values-d_169.html (visited on 01/01/2026).

[50] *Nitrogen price index*. 2025. url: https://businessanalytiq.com/procurementanalytics/index/nitrogen-price-index/ (visited on 09/12/2025).

[51] *Green Ammonia Price Trend, Index and Forecast*. 2025. url: https://www.imarcgroup.com/green-ammonia-price-trend (visited on 09/12/2025).

[52] *NKT Wins €650M HVDC Cable Deal for Bornholm Energy Island*. 8th Sept. 2025. url: https://www.sa-world.com/blog/nkt-wins-eur650m-hvdc-cable-deal-for-bornholm-energy-island.html (visited on 09/12/2025).

[53] Ministry, D. E. *Finding your way cheapest way to a low carbon future - The Danish Levelized Cost of Energy Calculator*. Tech. rep. Danish Energy Ministry, 2018. url: https://www.bp.com/content/dam/bp/business-sites/en/global/corporate/pdfs/energy-economics/energy-outlook/bp-energy-outlook-2025.pdf (visited on 09/12/2025).

[54] Pawlowski, R. *The Value of Place: Assessing Location-Based Economic Feasibility of Data Center Heat Recovery*. Aalborg University Copenhagen, 2025.

[55] Forum, E. I. *Making Energy Islands a Success: Roadmap for Offshore Energy Hub Development*. 2025.

[56] *Project Management Body of Knowledge*. Project Management Institute, 2000.

[57] Stenberg, C., Stottrup, J. G. et al. 'Long-term effects of an offshore wind farm in the North Sea on fish communities'. In: *Marine Ecology Progress Series* (2015).

[58] Degraer, S. et al. 'Offshore Wind Farm Artificial Reefs Affect Ecosystem Structure and Functioning: A Synthesis'. In: *Oceanography* (2020).

10 Appendix

10.1 Interview IVA

Everyone says hi.

Raphaël: We are conducting a project on offshore energy hubs. We have chosen to focus on the Bornholm Energy Island, as it already has some approved projects and others not yet approved. We wanted to talk with you to gather information and make sure we are looking in the right direction.

Iva: Can we have a round of introductions, just to hear what your backgrounds are and what you have been working on so far?

Everyone introduces themselves.

Iva: OK, great. Thanks. So now I have a picture of where you are coming from. How can I help you today?

Marcelina: We prepared some questions. For now, we have conducted research and a literature review, and we have started writing the basic structure of our report. But we stopped before the simulation and calculation parts. Our project focuses mostly on the economic and technological aspects, and we are not really looking into the social or spatial elements. That is not our main focus. We also have not yet been introduced to any of the software such as EnergyPRO, which we will use for the simulations.

We are wondering about some of the aspects we have already worked on, and also about the future calculations we plan to perform.

Our main topic is offshore energy hubs, and we are wondering whether, for Bornholm, considering only the currently approved projects, which involve offshore wind turbines and the

cable connections to Zealand and Germany, it is still accurate to call it an "energy island." In the literature, an offshore energy island usually involves generation, transmission, but also storage and conversion. If we are only talking about wind turbines producing electricity, we only have input and output. So we are wondering whether our research question and terminology are appropriate, whether we can call Bornholm an offshore energy island under the current plans.

Iva: Yes, that makes sense. The reason it is still referred to as Bornholm Energy Island is that originally there were plans to build a Power-to-X facility close to the connection point on Bornholm. But due to the delay and challenges in the first offshore wind tender round, many of the related developments stalled. So the name remained, but the follow-up did not progress as planned. It is interesting that you noticed this. The original plan had both wind generation and Power-to-X. To be honest, I do not know the exact current status of the Power-to-X project.

Marcelina: Yes, we read articles mentioning Power-to-X, and we are planning to include it together with ship fueling in one of our scenarios. Our baseline scenario will include only the approved wind turbines and the grid connection. Then, in our expansion scenarios, we plan to add Power-to-X and additional wind capacity, which would align more clearly with the definition of an energy island. We also want to look into the technological and economic benefits of creating such an offshore energy hub rather than a traditional offshore wind farm.

Iva: The original idea of Bornholm Energy Island was to showcase that this type of infrastructure is possible, the idea of placing large-scale offshore energy facilities close to an island, which then functions as an energy hub. It was intended as a demonstration project before pursuing the larger North Sea energy is-

land. This type of infrastructure will be important in the future, but it is also a significant financial commitment.

Bornholm has been seen as a potential export point for wind power, especially due to the connection with Germany. However, this created debate locally, because many residents feel that the island mostly serves national export interests without receiving sufficient benefits. I supervised two student projects on this. One report is publicly available, it focuses on stakeholder views and local perceptions. Another project used EnergyPLAN and MultiNode to evaluate whether exporting to Poland or Sweden in addition to Germany is beneficial. That report is currently confidential, but I can check whether it can be shared, because it provides a good overview of who benefits from the Bornholm Energy Island project.

Marcelina: Yes, thank you. Our initial idea was also to consider potential expansion with a connection to Poland, but we later decided to focus more closely on wind expansion and Power-to-X. But having an understanding of the results from those studies would be very helpful for our report.

Iva: I will check on that. The main conclusion in the confidential study was that exporting primarily benefits Germany, whereas Denmark benefits less directly. But I will try to confirm whether sharing the project is possible.

Marcelina: OK. Regarding Power-to-X, do you think Power-to-X is the right direction for Bornholm, or are other technologies such as large-scale storage or hydrogen export better for expansion?

Iva: Power-to-X fits well with Bornholm's ambitions to become a fueling hub for the Baltic Sea. However, most storage solutions currently tested in Denmark are small-scale, whereas this would be quite large. There has also been debate about whether

the local community could benefit from excess heat from the Power-to-X facility, for example by integrating it into district heating. The choice of fuel is also important: methanol production requires CO_2, and there may not be sufficient CO_2 resources locally, so ammonia is more realistic. There have also been discussions about producing hydrogen and exporting it instead.

Marcelina: Yes, we were mostly considering ammonia production. From your perspective, do you think producing ammonia for fueling ships is better than producing hydrogen and exporting it to mainland networks?

Iva: Based on the expected maritime fuel corridors, ammonia likely makes more sense, because it is easier to transport and aligns with emerging shipping infrastructure. However, if hydrogen ferries are introduced in the region, local hydrogen use could also become important.

Marcelina: Yes, thank you. We know that you are involved in the Nordic Hydrogen Valleys / Hubs project. What lessons from that project could apply to Bornholm?

Iva: The Nordic Hydrogen Valleys project is not only analysing existing hubs but also identifying where new hydrogen production hubs could emerge based on future demand. Many hubs are currently small-scale, whereas Bornholm could become a larger example, but only if the Power-to-X facility is included. One key lesson is that hydrogen hubs only work if there are nearby off-takers who will actually use the produced fuel. So if Bornholm expands into Power-to-X, ensuring strong demand is essential. I can share the project webpage, and next year, results will be published that may be relevant for your report.

Marcelina: OK, perfect. For our last question, we are not yet familiar with

EnergyPLAN or EnergyPRO. Raphaël has started exploring EnergyPRO for our project. We plan to analyse three scenarios: (1) the baseline with wind turbines and grid connections, (2) an expansion scenario with Power-to-X and ammonia production for ship fueling, and (3) an expanded wind capacity scenario. Do you think EnergyPRO is suitable for conducting both economic and technological simulations for these scenarios?

Iva: Yes, EnergyPRO is suitable for that. However, keep in mind that similar studies have been conducted in previous student projects. You should review those first so you can clearly define what new insights your project contributes. I will send you the two relevant reports once I verify access.

Marcelina: Thank you very much for your time and for answering all of our questions. We feel more confident now that we are moving in the right direction. We may contact you again if further questions come up.

Iva: You are welcome. Just reach out if you need anything else. Good luck with your project.

Everyone: Bye.

10.2 Power to X

10.2.1 PEMEC electrolysis

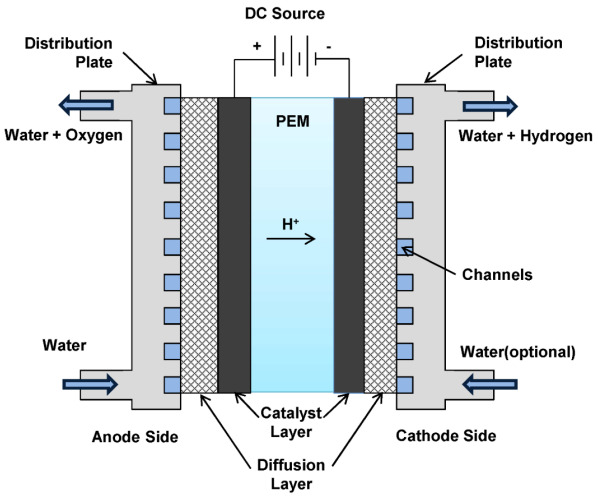


Figure 26: Schematic of a PEMEC cell (source: [38])

Figure 26 illustrates the fundamental operating principle of a Proton Exchange Membrane electrolyser. A PEMEC is composed of two half-cells, separated by a thin proton exchange membrane. The electrochemical reactions occurring at the anode and cathode, as well as the overall cell reaction, are expressed as follows:

$$\text{Anode:} \quad \mathrm{H_2O} \rightarrow 2\mathrm{H^+} + \tfrac{1}{2}\mathrm{O_2} + 2\mathrm{e^-}$$

$$\text{Cathode:} \quad 2\mathrm{H^+} + 2\mathrm{e^-} \rightarrow \mathrm{H_2}$$

$$\text{Overall:} \quad \mathrm{H_2O(l)} \rightarrow \mathrm{H_2(g)} + \tfrac{1}{2}\mathrm{O_2(g)}$$

These equations describe the water electrolysis process, where water is decomposed into hydrogen and oxygen gases through the application of an electric current [38].